

The Riemannian Geometry of Financial Correlation

Geodesic Distance, Estimation Quality,
and the Detection of Systemic Stress

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1 Abstract

This paper develops the Riemannian geometric framework for the space of correlation matrices $\text{Corr}(n)$ and investigates whether the ambient affine-invariant distance inherited from $\text{SPD}(n)$ provides a more informative measure of correlation stress (defined as a deviation of a market correlation matrix from a baseline) when compared to Euclidean or log-Euclidean distance. The theoretical chapters establish $\text{Corr}(n)$ as an embedded submanifold of $\text{SPD}(n)$, describe and derive the ambient geodesic distance formula, and show formally that the sensitivity of this distance to estimation noise depends on the smallest eigenvalues of the correlation matrix, showing that there exists a relationship between geometric richness and statistical stability. These chapters are also supported by figures, intuition, and motivating explanations so that the paper may serve as a pedagogical resource as well as a research investigation.

The empirical chapters then examine seven out-of-sample tests applied to 93 large-capitalisation US equities over the period 2003–2023: a controlled Monte Carlo experiment, regime classification, risk forecasting, correlation forecast accuracy, Granger causality analysis, a mechanical defensive backtest, and a precision–recall regime detector. Each test is conducted across five estimation methods: rolling windows (both raw and denoised through eigenvalue clipping) and three EWMA half-life configurations, to assess whether any observed advantage of the geodesic metric is robust across estimation regimes, or instead driven by the choice of estimator.

The central hypothesis is that the geodesic metric should outperform Euclidean distance when the correlation estimate is well-conditioned, but underperform when estimation noise dominates the smaller eigenvalues. This is tested across all seven tasks, with all results reported, including those where the geodesic metric does not perform best. Importantly, the full experimental setup—tests, estimators, and metrics—was specified in advance, rather than chosen after seeing the data, to avoid p-hacking or selectively highlighting favourable outcomes.

2 Introduction

The relationships between financial assets evolve over time, reflecting changing market conditions [1]. These changes are particularly pronounced during periods of market stress, when diversification benefits deteriorate [2] and the market tends towards unilateral movements [3, 4]. A key challenge is therefore how to measure shifts in dependence in a principled way.

This problem is typically approached through correlation matrices, which provide an intuitive representation of pairwise dependencies between asset returns [5]. The appropriate question to ask then is: what is the best way to measure how much a correlation matrix has moved?

The naive answer is to use the Euclidean distance, by treating the matrix as a vector of entries and measuring the straight-line distance between them in the ambient space. This is simple, computationally cheap and appears at a glance to be a reasonable strategy, however it ignores a key fact about that ambient space [6]: correlation matrices do not in fact live in a flat Euclidean space. They live in $\text{Corr}(n)$, the set of symmetric positive definite matrices with unit diagonal, which is a

curved Riemannian manifold [7]. Measuring distances on a curved surface with a flat ruler can give incorrect results, and as this paper argues, in periods of market stress this error can be substantial.

To build intuition, consider the simplest case $n = 1$, for which $\text{SPD}(1) = \mathbb{R}_{>0}$. In this setting, matrices reduce to positive scalars, and a congruence transformation $P \mapsto GPG^T$ corresponds simply to a rescaling by G^2 , which can be interpreted as a change of units.

Take $P = [1]$ and $Q = [4]$, and apply the rescaling $G = [2]$, giving $P' = [4]$ and $Q' = [16]$. Under the Euclidean metric, the distance changes from $|4 - 1| = 3$ to $|16 - 4| = 12$. Although the underlying relationship between P and Q has not changed, the distance increases by a factor of four purely due to a change of scale.

It would therefore be beneficial to identify a particular notion of distance that is invariant under such transformations. If the distance depends on the choice of units, then it does not capture the underlying geometric properties effectively.

The affine-invariant metric alleviates this concern by measuring change multiplicatively rather than additively [8]. In this case, the distance is given by $|\log(Q/P)| = |\log 4|$, which remains unchanged under the rescaling. This invariance arises due to the fact that the ratio Q/P is all that is relied upon, and is therefore also independent of the choice of units.

To improve rigor in the presented arguments, I include and explore an intermediate between Euclidean and the affine-invariant metric: the *log-Euclidean* distance, $d_{LE}(P, Q) = \|\log P - \log Q\|_F$ [7], which also yields $|\log 4|$ in the case P, Q are both scalar. However, this equivalence no longer holds in higher dimensions. Specifically, for $n \geq 2$, multiplication of matrices becomes non-commutative, so the identity $\log(P^{-\frac{1}{2}}QP^{-\frac{1}{2}}) = \log Q - \log P$ no longer holds in general. Therefore, the log-Euclidean and affine-invariant metrics capture unique notions of change: the log-Euclidean compares matrices by comparing them in a logarithmic coordinate system, whereas the affine-invariant metric measures changes relative to the geometry of the base points themselves.

Including both metrics in the empirical analysis allows for the opportunity to consider two questions. The first is whether working in the logarithmic domain alone improves the stability of computations, and the second is whether the extra understanding of geometric structure imposed by the affine invariance proves to be any more useful.

Beyond the choice of metrics in computing differences, there are other concerns to consider. Correlation matrices are estimated from finite samples, and the quality of that estimate can greatly impact the computations of distance [5]. The affine-invariant metric involves the matrix inverse P^{-1} and the matrix logarithm $\log P$, both of which greatly amplify small eigenvalues [9]. If computations are performed using a noisy estimate, the smallest eigenvalues are dominated by estimation error, resulting in the geodesic metric measuring noise rather than genuine structural change. This paper shows empirically that the estimation method matters at least as much as the metric, and identifies the specific estimation conditions discovered under which the geodesic metric provides meaningful value.

The paper is presented in three main sections: the mathematical theory, the empirical tests and results, and the discussions. The remainder of this chapter reviews the relevant literature and

positions this paper within that context. In chapters 3 and 4, I develop the Riemannian geometry of $\text{SPD}(n)$: the manifold structure, the affine-invariant metric, geodesics, and the distance formula. Chapter 5 establishes $\text{Corr}(n)$ as an embedded submanifold, and makes explicit an important qualification to the paper’s geometric interpretation: the distance we compute is the *ambient* $\text{SPD}(n)$ affine-invariant distance evaluated at points of $\text{Corr}(n)$, not the intrinsic geodesic distance on $\text{Corr}(n)$ itself. In other words, although the endpoints P, Q live in $\text{Corr}(n)$, we “leave” $\text{Corr}(n)$ by calculating distance through the parent space $\text{SPD}(n)$, rather than restricting the paths to $\text{SPD}(n)$. This process is standard in the applied literature [10] and mathematically sound, but it is a weaker object than the intrinsic geodesic. Despite this, it is important to be explicit. In chapter 6, I develop the theory of correlation matrix estimation, covering sample correlation matrices, eigenvalue clipping via the Marchenko–Pastur law [11], and EWMA estimation. There is a particular focus on how each of these methods interacts with the three distance metrics previously described. Chapter 7 defines the seven statistical tests I chose in the empirical analysis. Chapter 8 presents the results, Chapter 9 discusses the findings and Chapter 10 provides concluding remarks.

2.1 Related Work

A differential geometric approach to statistical analysis was first utilised by Mahalanobis [12], and later, the affine-invariant metric used in this paper was first introduced by Siegel [13], within the context of symmetric spaces. This was later developed as a Riemannian framework by Pennec, Fillard, and Ayache [14], who established the exponential and logarithmic maps, geodesic formulas, and statistical tools (Fréchet means, principal geodesic analysis) on $\text{SPD}(n)$. Bhatia’s monograph [8] provided comprehensive insights into the geometry of positive definite matrices, including the affine-invariant metric, the log-Euclidean metric, and their properties. The log-Euclidean framework was developed by Arsigny, Fillard, Pennec, and Ayache [7], who showed that the log map provides a global diffeomorphism from $\text{SPD}(n)$ to $\text{Sym}(n)$, endowing $\text{SPD}(n)$ with a flat Riemannian structure that is computationally simpler than the affine-invariant metric, yet still respects the positive definite constraint. Dryden, Koloydenko, and Zhou [15] developed non-Euclidean statistical methods for covariance matrices, including Procrustes-type distances and power-Euclidean metrics that provide a family interpolating between Euclidean and log-Euclidean structures.

On the estimation side, the Marchenko–Pastur law [11] provides the foundational result on the eigenvalue distribution of sample covariance matrices in the high-dimensional regime. Ledoit and Wolf [16, 17] developed linear and nonlinear shrinkage estimators that systematically correct for the finite-sample distortion of sample eigenvalues. The EWMA covariance estimator was popularised by the RiskMetrics methodology [18], and Engle [19] extends EWMA to allow time-varying correlation dynamics through maximum likelihood estimation. The developed model is known as the Dynamic Conditional Correlation (DCC) model.

In the financial applications literature, correlation-based measures of systemic risk include the absorption ratio of Kritzman, Li, Page, and Rigobon [20], which measures the fraction of total variance explained by a fixed number of eigenvectors, and the principal components-based inter-

connectedness measures of Billio, Getmansky, Lo, and Pelizzon [21]. These approaches share the premise that the eigenstructure of the correlation matrix contains information about systemic risk, but they measure it through spectral statistics rather than geometric distance.

The application of Riemannian geometry specifically to financial covariance or correlation matrices is more recent and less developed. To the best of my knowledge, no prior work has specifically conducted a controlled, multi-test empirical comparison of the affine-invariant, log-Euclidean, and Euclidean distances on financial correlation matrices across multiple estimation methods with out-of-sample evaluation. As such, the contribution of this paper is to provide this comparison, and to show that the interaction between estimation quality and metric choice is a key determinant of practical performance.

3 Geometry of $\text{Sym}(n)$ and $\text{SPD}(n)$

We begin the theory by establishing the geometric objects and structures that will act as the foundations for the rest of the paper. The goal of Chapter 3 is to show that $\text{SPD}(n)$ carries a natural Riemannian metric that is invariant under congruence transformations and therefore provides a geometrically meaningful notion of distance.

Definition 3.1. *The space of $n \times n$ real symmetric matrices is*

$$\text{Sym}(n) = \{A \in \mathbb{R}^{n \times n} : A = A^\top\}. \quad (1)$$

and serves as our initial ambient space.

Definition 3.2. *The space of $n \times n$ symmetric positive definite matrices is*

$$\text{SPD}(n) = \{P \in \text{Sym}(n) : v^\top P v > 0 \ \forall v \in \mathbb{R}^n \setminus \{0\}\}. \quad (2)$$

and is a subset of $\text{Sym}(n)$.

Remark 3.1. *The space $\text{Sym}(n)$ is isomorphic to $\mathbb{R}^{n(n+1)/2}$ as a vector space, since a symmetric matrix is determined entirely by its upper triangular entries. In particular, $\text{Sym}(n)$ is a smooth manifold of dimension $n(n+1)/2$.*

With this remark, it becomes clear that we can suitably treat $\text{Sym}(n)$ as a vector space, and hence a smooth manifold. Now, we establish that $\text{SPD}(n)$ is an open subset of $\text{Sym}(n)$, as this allows for the inheritance of the structure of the smooth manifold, which we will later use to define our Riemannian metric.

Proposition 3.1. *$\text{SPD}(n)$ is an open subset of $\text{Sym}(n)$.*

Proof: Let $A \in \text{Sym}(n)$ with eigenvalues $\lambda_1(A), \dots, \lambda_n(A) \in \mathbb{R}$, which are real-valued since A is symmetric. Define the function

$$\lambda_{\min} : \text{Sym}(n) \rightarrow \mathbb{R}, \quad \lambda_{\min}(A) = \min\{\lambda_i(A) : i = 1, \dots, n\} \quad (3)$$

Since eigenvalues are roots of the characteristic polynomial $\det(A - \lambda I) = 0$, the eigenvalues of A vary continuously with the entries of A , and the minimum of finitely many continuous functions is itself continuous. Therefore $\lambda_{\min}$ is continuous.

Now, we claim that $\text{SPD}(n) = \lambda_{\min}^{-1}((0, \infty))$, or in other words that $A \in \text{SPD}(n)$ if and only if $\lambda_{\min}(A) > 0$.

($\Rightarrow$) Suppose $A \in \text{SPD}(n)$, so $v^T A v > 0$, $\forall v \in \mathbb{R}^n \setminus \{0\}$. Let $\lambda_i(A)$ be any eigenvalue of A with corresponding eigenvector $v_i \neq 0$. Then

$$v_i^T A v_i = v_i^T \lambda_i v_i = \lambda_i \|v_i\|^2 > 0. \quad (4)$$

Since $\|v_i\|^2 > 0$, we conclude $\lambda_i > 0$, and since this is true for all λ_i , then it must also be true for the minimum $\lambda_{\min}(A) > 0$.

($\Leftarrow$) Suppose now that $\lambda_{\min}(A) > 0$. Then all eigenvalues of A must be positive. Since A is symmetric, it is orthogonally diagonalisable and so can be written as $A = Q \Lambda Q^T$ where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ with each $\lambda_i > 0$. Then for any nonzero vector v

$$v^T A v = v^T Q \Lambda Q^T v = \sum_{i=1}^n \lambda_i (Q^T v)_i^2 > 0, \quad (5)$$

which holds since every $\lambda_i > 0$ and since Q is orthogonal with $v \neq 0$, we have $Q^T v \neq 0$, so at least one $(Q^T v)_i \neq 0$. Hence $A \in \text{SPD}(n)$.

Since the interval $(0, \infty)$ is open in $\mathbb{R}$ and $\lambda_{\min}$ is a continuous function, its preimage $\lambda_{\min}^{-1}((0, \infty))$ is open in $\text{Sym}(n)$. Therefore $\text{SPD}(n)$ is open in $\text{Sym}(n)$. $\square$

Remark 3.2. Since $\text{SPD}(n)$ is open in $\text{Sym}(n)$, the tangent space at any point $P \in \text{SPD}(n)$ is isomorphic to $\text{Sym}(n)$ itself, independent of any choices of basis. This follows as any curve $\gamma(t) = P + tS$ for $S \in \text{Sym}(n)$ is a valid smooth curve through P for sufficiently small t , giving $T_P \text{SPD}(n) \cong \text{Sym}(n)$.

Remark 3.3. For any $P \in \text{SPD}(n)$, the inverse P^{-1} exists and belongs to $\text{SPD}(n)$. Since all eigenvalues of P are positive, P must be invertible and P^{-1} is also symmetric with eigenvalues $\lambda_i^{-1} > 0$.

Remark 3.4. For any $P \in \text{SPD}(n)$ with spectral decomposition $P = Q \Lambda Q^T$, the matrix square root is defined as

$$P^{\frac{1}{2}} = Q \text{diag}(\lambda_1^{\frac{1}{2}}, \dots, \lambda_n^{\frac{1}{2}}) Q^T \quad (6)$$

and satisfies $P^{\frac{1}{2}} \in \text{SPD}(n)$. In particular, $P^{-\frac{1}{2}}$ exists and is also in $\text{SPD}(n)$.

Having identified the tangent space at each point, we now seek to equip $\text{SPD}(n)$ with a Riemannian metric. As shown in the Introduction, the Euclidean metric inherited from $\text{Sym}(n)$ is

not invariant under congruence transformations, $P \mapsto GPG^T$, so we instead require a metric that respects these symmetries. The affine invariant Riemannian metric is the natural choice.

3.1 The Affine-Invariant Riemannian Metric

The Euclidean inner product on $\text{Sym}(n)$, given by $\langle U, V \rangle = \text{tr}(UV)$, is not invariant under congruence transformations $P \mapsto GPG^T$, as demonstrated in the Introduction. We instead equip $\text{SPD}(n)$ with the affine-invariant metric.

Definition 3.3. For $P \in \text{SPD}(n)$ and $U, V \in T_P \text{SPD}(n) \cong \text{Sym}(n)$, the affine-invariant Riemannian metric is

$$\langle U, V \rangle_P = \text{tr}(P^{-1}UP^{-1}V). \quad (7)$$

It can be thought of as though P^{-1} is correcting for the fact that different points of $\text{SPD}(n)$ have different natural scales. A direction of movement of a fixed size should not mean the same thing everywhere: it is less significant when P is large and more significant when P is small. The metric accounts for this by measuring U relative to P , rather than in Euclidean terms.

Proposition 3.2. The affine invariant Riemannian metric $\langle U, V \rangle_P = \text{tr}(P^{-1}UP^{-1}V)$ defines a valid inner product on $T_P \text{SPD}(n)$ for each $P \in \text{SPD}(n)$.

Proof: We verify the three properties of an inner product.

Linearity:

$$\begin{aligned} \langle \alpha U + \beta V, W \rangle_P &= \text{tr}(P^{-1}(\alpha U + \beta V)P^{-1}W) \\ &= \alpha \text{tr}(P^{-1}UP^{-1}W) + \beta \text{tr}(P^{-1}VP^{-1}W) \\ &= \alpha \langle U, W \rangle_P + \beta \langle V, W \rangle_P \end{aligned}$$

Symmetry:

$$\langle U, V \rangle_P = \text{tr}(P^{-1}UP^{-1}V) = \text{tr}(P^{-1}VP^{-1}U) = \langle V, U \rangle_P \quad (8)$$

where we have used the cyclic property of the trace.

Positive definiteness: Let $U \in \text{Sym}(n)$ be nonzero. Then

$$\langle U, U \rangle_P = \text{tr}(P^{-1}UP^{-1}U) = \text{tr}\left((P^{-\frac{1}{2}}UP^{-\frac{1}{2}})^2\right) = \sum_i \lambda_i^2 \geq 0$$

where λ_i are the eigenvalues of the symmetric matrix $P^{-\frac{1}{2}}UP^{-\frac{1}{2}}$, which are real. This sum equals zero only if all $\lambda_i = 0$, or equivalently $P^{-\frac{1}{2}}UP^{-\frac{1}{2}} = 0$. Since $P^{-\frac{1}{2}}$ is invertible, this implies $U = 0$, a contradiction. Therefore $\langle U, U \rangle_P > 0$ for all nonzero U .

Since this is an inner product at each point P , the last step is to show that it also depends

smoothly on P , or in other words that

$$P \mapsto \text{tr}(P^{-1}UP^{-1}V) \quad (9)$$

is smooth for any arbitrarily chosen tangent vectors $U, V \in \text{Sym}(n)$. Since matrix inversion and the trace are both smooth operations, $\langle U, V \rangle_P$ varies smoothly with P , and therefore defines a Riemannian metric on $\text{SPD}(n)$. $\square$

Proposition 3.3. *For any $P \in \text{SPD}(n)$ and invertible $G \in \mathbb{R}^{n \times n}$, we have $GPG^\top \in \text{SPD}(n)$.*

Proof: Symmetry: $(GPG^\top)^\top = GPG^\top$. Positive definiteness: for nonzero x , $x^\top(GPG^\top)x = (G^\top x)^\top P(G^\top x) > 0$ since $G^\top x \neq 0$. $\square$

Proposition 3.4. *The affine-invariant metric is invariant under congruence transformations:*

$$\langle GUG^\top, GVG^\top \rangle_{GPG^\top} = \langle U, V \rangle_P. \quad (10)$$

Proof:

$$\begin{aligned} \langle GUG^\top, GVG^\top \rangle_{GPG^\top} &= \text{tr}\left((GPG^\top)^{-1}GUG^\top(GPG^\top)^{-1}GVG^\top\right) \\ &= \text{tr}\left(G^{-\top}P^{-1}UP^{-1}VG^\top\right) \\ &= \text{tr}\left(P^{-1}UP^{-1}V\right) \\ &= \langle U, V \rangle_P, \end{aligned}$$

using the cyclic property of trace. $\square$

With these results, the geometric structure of $\text{SPD}(n)$ is now fully established. We now turn our attention to the geodesic structure of the manifold which will provide us with our notion of distance between points in $\text{SPD}(n)$.

4 Geodesics and Distance on $\text{SPD}(n)$

To study distances on $\text{SPD}(n)$, we require the notion of a geodesic, which are curves that locally minimise distance. To outline our geodesic properties, we establish some fundamental definitions.

Definition 4.1. *The matrix exponential is defined by the same power series as the scalar exponential*

$$\exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots \quad (11)$$

which converges for any square matrix $A \in \mathbb{R}^{n \times n}$.

We need to be careful as the matrix logarithm is not defined for all matrices, so we first note the following:

Remark 4.1. The matrix logarithm is well defined on $SPD(n)$. For $P \in SPD(n)$ with spectral decomposition $P = Q\Lambda Q^T$, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, each $\log \lambda_i$ is a well defined real number since all eigenvalues satisfy $\lambda_i > 0$. Furthermore, $\log(P) \in \text{Sym}(n)$, meaning the logarithm maps back into the tangent space $T_P SPD(n) \cong \text{Sym}(n)$.

We can now therefore define the matrix logarithm on $SPD(n)$ as follows:

Definition 4.2. For $P \in SPD(n)$ with spectral decomposition $P = Q\Lambda Q^T$, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, the matrix logarithm is defined as

$$\log(P) = Q \text{diag}(\log \lambda_1, \dots, \log \lambda_n) Q^T \quad (12)$$

Now, with these definitions in hand, we define exponential and logarithmic maps, which will allow us to move between the manifold $SPD(n)$ and its tangent spaces $T_P SPD(n)$.

Definition 4.3. The exponential map $\text{Exp}_P : T_P SPD(n) \rightarrow SPD(n)$ is defined as

$$\text{Exp}_P(U) = P^{\frac{1}{2}} \exp(P^{-\frac{1}{2}} U P^{-\frac{1}{2}}) P^{\frac{1}{2}} \quad (13)$$

for $P \in SPD(n)$ and $U \in T_P SPD(n) \cong \text{Sym}(n)$.

The intuition for this is as follows:

- $P^{-\frac{1}{2}} U P^{-\frac{1}{2}}$ is the conjugation of the tangent vector U into the flat plane at the identity I .
- $\exp(\cdot)$ is the application of the matrix exponential in the flat world at I .
- $P^{\frac{1}{2}}(\cdot)P^{\frac{1}{2}}$ is the conjugation back to curved space at the point P .

Definition 4.4. The logarithmic map $\text{Log}_P : SPD(n) \rightarrow T_P SPD(n)$ is the inverse of the exponential map, defined as

$$\text{Log}_P(Q) = P^{\frac{1}{2}} \log(P^{-\frac{1}{2}} Q P^{-\frac{1}{2}}) P^{\frac{1}{2}} \quad (14)$$

for $P, Q \in SPD(n)$.

The intuition here is the reverse of the exponential:

- $P^{-\frac{1}{2}} Q P^{-\frac{1}{2}}$ is the conjugation of Q into the flat world at I .
- $\log(\cdot)$ is the application of the matrix logarithm in flat space.
- $P^{\frac{1}{2}}(\cdot)P^{\frac{1}{2}}$ is the conjugation back into the tangent space at point P .

The exponential and logarithmic maps provide a complete correspondence between directions in the tangent space and points on the manifold, with Exp_P tracing geodesics from P , and Log_P recovering the direction between two points. We now formally present the geodesic in two forms. The first represents the geodesic from a point P in the direction of the tangent vector U (tangent space into manifold), and the second represents the geodesic between two points P and Q (manifold into tangent space).

Definition 4.5. The geodesic on $SPD(n)$ passing through $P \in SPD(n)$ in the direction of $U \in T_P SPD(n) \cong Sym(n)$ is the curve $\gamma : [0, 1] \rightarrow SPD(n)$ defined by

$$\gamma(t) = Exp_P(tU) = P^{\frac{1}{2}} \exp\left(t \cdot P^{-\frac{1}{2}} U P^{-\frac{1}{2}}\right) P^{\frac{1}{2}} \quad (15)$$

The geodesic connecting P to $Q \in SPD(n)$ is obtained by setting $U = Log_P(Q)$, which gives

$$\gamma(t) = P^{\frac{1}{2}} \exp\left(t \cdot P^{-\frac{1}{2}} Log_P(Q) P^{-\frac{1}{2}}\right) P^{\frac{1}{2}} \quad (16)$$

with $\gamma(0) = P$ and $\gamma(1) = Q$.

4.1 The Geodesic Distance Formula

Definition 4.6. The Frobenius norm of a matrix $A \in \mathbb{R}^{n \times n}$ is defined as

$$\|A\|_F = \sqrt{tr(A^T A)} = \sqrt{\sum_{i,j} a_{ij}^2} \quad (17)$$

Proposition 4.1. The geodesic distance between two points $P, Q \in SPD(n)$ is given by

$$d(P, Q) = \left\| \log(P^{-\frac{1}{2}} Q P^{-\frac{1}{2}}) \right\|_F \quad (18)$$

Proof: The length of a curve $\gamma : [0, 1] \rightarrow SPD(n)$ on a Riemannian manifold is defined as

$$L(\gamma) = \int_0^1 \|\dot{\gamma}(t)\|_{\gamma(t)} dt \quad (19)$$

and the geodesic distance is the infimum of this over all curves connecting P to Q :

$$d(P, Q) = \inf_{\gamma} L(\gamma) \quad (20)$$

The geodesic connecting P to Q is

$$\gamma(t) = P^{\frac{1}{2}} \exp\left(t \cdot P^{-\frac{1}{2}} U P^{-\frac{1}{2}}\right) P^{\frac{1}{2}} \quad (21)$$

where $U = Log_P(Q)$. Since the geodesic has constant speed under the affine invariant metric, the length simplifies to

$$L(\gamma) = \|\dot{\gamma}(0)\|_P = \|U\|_P = \|Log_P(Q)\|_P \quad (22)$$

Now, to compute $\|\text{Log}_P(Q)\|_P$, we use the previous definition of the logarithmic map and the metric:

$$\begin{aligned}
\|\text{Log}_P(Q)\|_P^2 &= \langle \text{Log}_P(Q), \text{Log}_P(Q) \rangle_P \\
&= \text{tr} (P^{-1} \text{Log}_P(Q) P^{-1} \text{Log}_P(Q)) \\
&= \text{tr} \left(\left(P^{\frac{1}{2}} \text{Log}_P(Q) P^{-\frac{1}{2}} \right)^2 \right) \\
&= \text{tr} \left(\left(\log(P^{-\frac{1}{2}} Q P^{-\frac{1}{2}}) \right)^2 \right) \\
&= \left\| \log(P^{-\frac{1}{2}} Q P^{-\frac{1}{2}}) \right\|_F^2
\end{aligned}$$

where the last equality uses the fact that for any symmetric matrix S , $\text{tr}(S^2) = \|S\|_F^2$. Taking square roots yields

$$d(P, Q) = L(\gamma) = \left\| \log(P^{-\frac{1}{2}} Q P^{-\frac{1}{2}}) \right\|_F \quad (23)$$

□

4.2 The Log-Euclidean Distance

Definition 4.7. *The log-Euclidean distance between $P, Q \in \text{SPD}(n)$ is*

$$d_{LE}(P, Q) = \|\log P - \log Q\|_F. \quad (24)$$

The log-Euclidean metric is a natural intermediate between Euclidean and geodesic metrics. Similar to the geodesic metric, it operates in the logarithmic domain, however it is unique in that it does not conjugate by $P^{-1/2}$ before taking the logarithm, so it does not adapt to the local geometry at the base point.

Proposition 4.2. *The log-Euclidean distance is a valid metric on $\text{SPD}(n)$.*

Proof: The map $\log : \text{SPD}(n) \rightarrow \text{Sym}(n)$ is a bijection (with inverse $\exp$). For any $P, Q, R \in \text{SPD}(n)$:

Non-negativity and identity: $d_{LE}(P, Q) = \|\log P - \log Q\|_F \geq 0$, with equality if and only if $\log P = \log Q$, which holds if and only if $P = Q$ since $\log$ is injective.

Symmetry:

$$\begin{aligned}
d_{LE}(P, Q) &= \|\log P - \log Q\|_F \\
&= \|\log Q - \log P\|_F \\
&= d_{LE}(Q, P).
\end{aligned} \quad (25)$$

Triangle inequality: Since the Frobenius norm satisfies the triangle inequality on $\text{Sym}(n)$,

$$\begin{aligned}
d_{LE}(P, R) &= \|\log P - \log R\|_F \\
&\leq \|\log P - \log Q\|_F + \|\log Q - \log R\|_F \\
&= d_{LE}(P, Q) + d_{LE}(Q, R).
\end{aligned} \tag{26}$$

More precisely, d_{LE} is what is known as the pullback of the Frobenius distance on $\text{Sym}(n)$ through the bijection $\log$, therefore inheriting all metric properties. $\square$

Remark 4.2. *The log map identifies $\text{SPD}(n)$ with $\text{Sym}(n) \cong \mathbb{R}^{n(n+1)/2}$, which is flat. The log-Euclidean metric therefore corresponds to a flat Riemannian structure on $\text{SPD}(n)$, in contrast to the affine-invariant metric which induces a space of non-positive sectional curvature [8].*

Proposition 4.3. *For $n = 1$, the geodesic and log-Euclidean distances coincide:*

$$d_g(p, q) = d_{LE}(p, q) = |\log(q/p)| \text{ for } p, q > 0. \tag{27}$$

Proof:

$$\begin{aligned}
&\text{For scalars, } p^{-\frac{1}{2}}qp^{-\frac{1}{2}} = \frac{q}{p}, \text{ so } d_g = \left| \log \left(\frac{q}{p} \right) \right|. \\
&\text{Also } \log p - \log q = -\log \left(\frac{q}{p} \right), \text{ so } d_{LE} = \left| \log \left(\frac{q}{p} \right) \right|.
\end{aligned} \tag{28}$$

$\square$

Example 4.1. *For $n = 2$, the two metrics diverge when the matrices do not commute. Let*

$$P = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}, \quad Q = \begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix}. \tag{29}$$

Both are symmetric positive definite (eigenvalues of P are $\{3.62, 1.38\}$, of Q are $\{4.24, 1.76\}$), but they do not commute: $PQ \neq QP$ because the off-diagonal entries have opposite signs, so the eigenvectors differ. Computing numerically: $d_g(P, Q) \approx 1.43$ and $d_{LE}(P, Q) \approx 1.31$.

The divergence arises because the geodesic distance computes $\log(P^{-1/2}QP^{-1/2})$, which conjugates Q into the frame of P before taking the logarithm, while the log-Euclidean distance computes $\log Q - \log P$ directly. For non-commuting matrices, the Baker–Campbell–Hausdorff formula [7] implies $\log(P^{-1/2}QP^{-1/2}) \neq \log Q - \log P$, and the discrepancy grows with the degree of non-commutativity.

Definition 4.8. *The Euclidean distance (Frobenius distance) between $P, Q \in \text{SPD}(n)$ is*

$$d_E(P, Q) = \|P - Q\|_F. \tag{30}$$

We now have three distance measures on $\text{SPD}(n)$, ordered by the amount of geometric structure they respect: Euclidean, log-Euclidean, and geodesic. All that is left before the empirical analysis is to explore the direct geometry of the space of correlation matrices $\text{Corr}(n)$.

5 Geometry of $\text{Corr}(n)$

Definition 5.1. *The set of $n \times n$ correlation matrices is defined as*

$$\text{Corr}(n) = \{A \in \text{SPD}(n) : A_{ii} = 1 \ \forall i = 1, \dots, n\} \quad (31)$$

That is, $\text{Corr}(n)$ consists of all symmetric positive definite matrices with ones along the diagonal.

The set $\text{Corr}(n)$ is by definition a subset of $\text{SPD}(n)$, but it does not trivially inherit a smooth manifold structure. Much of the research on this topic utilises the Lie group action on $\text{Corr}(n)$ (see for example [10]), however I will detail a proof without adding additional theory here. I instead apply the Regular Level Set Theorem, specifically describing $\text{Corr}(n)$ as a regular level set of a smooth map defined on $\text{SPD}(n)$. From this, it follows that $\text{Corr}(n)$ is an embedded submanifold of $\text{SPD}(n)$.

The Regular Level Set Theorem says that if we have a smooth map f and a regular value c , then the preimage $f^{-1}(c)$ is a smooth manifold. The goal of the proof then is simply to find a smooth map whose preimage at some point is exactly $\text{Corr}(n)$.

Proposition 5.1. *$\text{Corr}(n)$ is an embedded submanifold of $\text{SPD}(n)$ of dimension $\frac{n(n+1)}{2} - n$.*

Proof: Define the map $f : \text{SPD}(n) \rightarrow \mathbb{R}^n$ by

$$f(P) = (P_{11}, P_{22}, \dots, P_{nn}). \quad (32)$$

Then

$$f^{-1}(\mathbf{1}) = \{P \in \text{SPD}(n) : f(P) = \mathbf{1}\}, \quad (33)$$

where $\mathbf{1} = (1, 1, \dots, 1) \in \mathbb{R}^n$.

Since $f(P) = \mathbf{1}$ exactly when $P_{11} = \dots = P_{nn} = 1$, it follows that

$$f^{-1}(\mathbf{1}) = \text{Corr}(n). \quad (34)$$

We have found an appropriate mapping, and now it remains to verify that $\mathbf{1}$ is a regular value of f , which requires showing that

- f is smooth
- df_P is surjective at every $P \in \text{Corr}(n)$

Smoothness of f : Each component $f_i : \text{SPD}(n) \rightarrow \mathbb{R}$ is defined by $f_i(P) = P_{ii}$, which is just a coordinate projection on the vector space $\text{Sym}(n) \cong \mathbb{R}^{n(n+1)/2}$ shown in Remark 2.2. Coordinate projections are smooth, and since each component f_i is smooth, f is smooth.

Surjectivity of df_P : To apply the Regular Level Set Theorem, we must show that the derivative $df_P : T_P\text{SPD}(n) \rightarrow \mathbb{R}^n$ is surjective at every $P \in \text{Corr}(n)$. Since $\text{SPD}(n)$ is open in $\text{Sym}(n)$, we have $T_P\text{SPD}(n) \cong \text{Sym}(n)$, and the derivative of f at P in the direction of some tangent vector $S \in \text{Sym}(n)$ is

$$df_P(S) = \left. \frac{d}{dt} \right|_{t=0} f(P + tS) = \left. \frac{d}{dt} \right|_{t=0} (P_{11} + tS_{11}, \dots, P_{nn} + tS_{nn}) = (S_{11}, \dots, S_{nn}) \quad (35)$$

The derivative df_P measures how f changes as we move away from P , and since f only reads off diagonal entries, only the diagonal entries of the direction S matter.

To show df_P is surjective, let $(v_1, \dots, v_n) \in \mathbb{R}^n$ be arbitrarily chosen. We need to find $S \in \text{Sym}(n)$ such that $df_P(S) = (v_1, \dots, v_n)$. Taking $S = \text{diag}(v_1, \dots, v_n)$, which is symmetric since all off-diagonal entries are zeros, so we have

$$df_P(S) = (S_{11}, \dots, S_{nn}) = (v_1, \dots, v_n) \quad (36)$$

Since $(v_1, \dots, v_n)$ was arbitrarily chosen, df_P is surjective at every $P \in \text{Corr}(n)$.

By the Regular Level Set Theorem, since f is smooth and df_P is surjective at every point of $f^{-1}(\mathbf{1})$, the preimage $\text{Corr}(n) = f^{-1}(\mathbf{1})$ is an embedded submanifold of $\text{SPD}(n)$, and the dimension of $\text{Corr}(n)$ is

$$\dim(\text{Corr}(n)) = \dim(\text{SPD}(n)) - \dim(\mathbb{R}^n) = \frac{n(n+1)}{2} - n = \frac{n(n-1)}{2} \quad (37)$$

where we subtract n because the map f imposes n independent constraints (each of the n diagonal entries must be 1). $\square$

Having established that $\text{Corr}(n)$ is in fact an embedded submanifold of $\text{SPD}(n)$, we now identify its tangent space. Since $\text{Corr}(n)$ is a regular level set of f , its tangent space at each point is the kernel of df_P , which is the set of directions for which the diagonal entries do not change.

Proposition 5.2. *The tangent space of $\text{Corr}(n)$ at any point $P \in \text{Corr}(n)$ is*

$$T_P\text{Corr}(n) = \ker(df_P) = \{S \in \text{Sym}(n) : S_{ii} = 0, i = 1, \dots, n\} \quad (38)$$

Proof: Since $\text{Corr}(n) = f^{-1}(\mathbf{1})$ is a regular level set of f , the tangent space at any point $P \in$

$\text{Corr}(n)$ is the kernel of the derivative df_P . Recalling from our previous proof that

$$df_P(S) = (S_{11}, \dots, S_{nn}) \quad (39)$$

so a tangent vector $S \in \text{Sym}(n)$ lies in $\ker(df_P)$ if and only if $df_P(S) = \mathbf{0}$, or in other words if and only if $S_{ii} = 0$ for all $i = 1, \dots, n$. Therefore

$$T_P \text{Corr}(n) = \ker(df_P) = \{S \in \text{Sym}(n) : S_{ii} = 0, i = 1, \dots, n\} \quad (40)$$

In other words, $T_P \text{Corr}(n)$ is the set of $\text{Sym}(n)$ matrices with zeros on their diagonal. $\square$

With the tangent space identified, the affine invariant metric on $\text{SPD}(n)$ from Definition 3.3 (7) naturally restricts to $\text{Corr}(n)$.

5.1 Ambient versus Intrinsic Distance

For the purposes of this paper there requires a subtle yet important distinction to make. Given $P, Q \in \text{Corr}(n)$, the $\text{SPD}(n)$ geodesic connecting them does not in general remain inside $\text{Corr}(n)$. The intermediate points $\gamma(t)$ for $0 < t < 1$ may have diagonal entries that differ from one, since the unit-diagonal constraint is not preserved along the ambient geodesic. This means that the distance formula $d_g(P, Q) = \|\log(P^{-1/2}QP^{-1/2})\|_F$ is not the intrinsic geodesic distance on $\text{Corr}(n)$, but instead the ambient $\text{SPD}(n)$ distance evaluated at points P, Q of $\text{Corr}(n)$.

Since any curve in $\text{Corr}(n)$ is also a curve in $\text{SPD}(n)$, the ambient distance provides a lower bound:

$$d_g^{\text{SPD}}(P, Q) \leq d_g^{\text{Corr}}(P, Q). \quad (41)$$

Throughout this paper, we compute the ambient $\text{SPD}(n)$ distance. This is standard in the applied literature [14, 15, 10] for three reasons. First, it provides us with a closed-form expression, whereas the intrinsic $\text{Corr}(n)$ geodesic requires solving a constrained optimisation with no known closed-form for general n . Second, it is smooth and well defined everywhere on $\text{Corr}(n)$. Third, for our empirical purposes in later chapters, what matters is whether the distance captures meaningful variation, not whether it coincides exactly with the intrinsic length. Therefore we will continue the analysis with the ambient metric.

Before considering the general case, it is worth examining the most standard nontrivial setting. For $n = 2$, the entire correlation structure reduces to a single parameter ρ , which means the geodesic distance can be written in closed form as an explicit function of two real numbers. This makes it possible to directly observe how the geometry of $\text{Corr}(n)$ behaves, and in particular how the manifold stretches near its boundary. It is also worth noting that 2×2 correlation matrices commute, so this is the one case in which the log-Euclidean and geodesic distances match together; the divergence between the two metrics only emerges in higher dimensions.

Example 5.1. For $n = 2$, $\text{Corr}(2)$ is parameterised by a single correlation coefficient $\rho \in (-1, 1)$:

$$C(\rho) = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}. \quad (42)$$

The eigenvalues are $1 + \rho$ and $1 - \rho$. To compute $d_g(C(\rho_1), C(\rho_2))$, we can note that both matrices share the eigenvectors $\frac{1}{\sqrt{2}}(1, 1)^\top$ and $\frac{1}{\sqrt{2}}(1, -1)^\top$, so they commute. The matrix $C(\rho_1)^{-\frac{1}{2}}C(\rho_2)C(\rho_1)^{-\frac{1}{2}}$ has the eigenvalues

$$\frac{1 + \rho_2}{1 + \rho_1} \quad \text{and} \quad \frac{1 - \rho_2}{1 - \rho_1}. \quad (43)$$

Applying the logarithm and Frobenius norm:

$$d_g(C(\rho_1), C(\rho_2)) = \sqrt{\log^2\left(\frac{1 + \rho_2}{1 + \rho_1}\right) + \log^2\left(\frac{1 - \rho_2}{1 - \rho_1}\right)}. \quad (44)$$

It is important to note how the distance diverges as either matrix approaches a singularity ($\rho \rightarrow \pm 1$), since the logarithm of a ratio approaching zero or infinity diverges. This stretching near the boundary is geometrically reasonable; matrices near singularity occupy extreme positions on the manifold. This results in a behaviour which is meaningful in a financial context where near-perfect correlation represents a fundamentally different market movement.

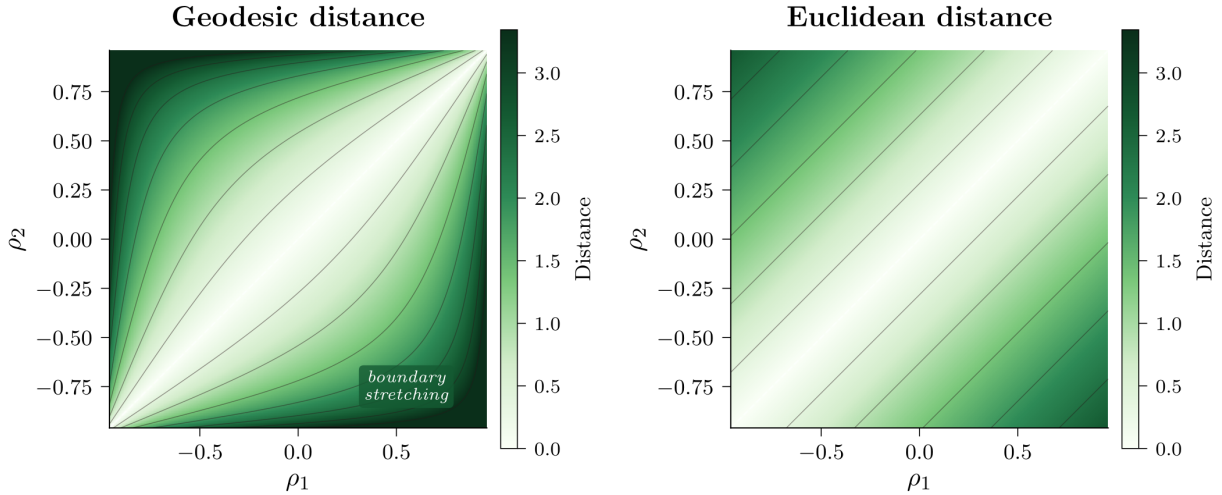


Figure 1: Distance between 2×2 correlation matrices $C(\rho_1)$ and $C(\rho_2)$, where ρ_i is the single off-diagonal correlation coefficient parameterising each matrix. Left: geodesic distance, showing pronounced stretching near the boundaries $\rho \rightarrow \pm 1$ where the matrices approach singularity. Right: Euclidean distance, which grows uniformly. Both panels share the same colour scale.

The results from Chapter 5 have shown that $\text{Corr}(n)$ is a well-defined Riemannian manifold, particularly an embedded submanifold of $\text{SPD}(n)$ with dimension $\frac{n(n-1)}{2}$ and is equipped with the affine invariant metric. This gives us a complete framework for measuring distances between corre-

lation matrices. However, the sample correlation matrices we observe are finite-sample estimators of the true underlying correlation structure, and the spectra of eigenvalues contain noise from estimation. Since our affine invariant metric involves P^{-1} and $\log(P)$, which are both sensitive to small eigenvalues, the resulting noise can destabilise geodesic computations and potentially distort any distances that we measure. To alleviate these concerns, the tools required to denoise the sample correlation matrix will be explored in the following chapter alongside the theory of estimation itself.

6 Correlation Matrix Estimation

In practical applications, correlation matrices are estimated from a finite sample of T observations of n elements. This finite sample introduces a core issue, however. The observed eigenvalue spectrum of C is not the same as the eigenvalue spectrum of the underlying true correlation matrix Σ . Some eigenvalues are inflated or deflated from the sample's noise, but the important notion is that this distortion is not random but systematic.

In our circumstances, both P^{-1} and $\log(P)$ terms within our Riemannian metric amplify small eigenvalues, and as a result may produce signals that reflect estimation error rather than meaningful correlation. In this section I introduce some standard tools from Random Matrix Theory which help to separate the signal eigenvalues from the noise eigenvalues, but first we must define what a sample correlation matrix is.

6.1 Sample Correlation Matrices and the Concentration Ratio

Definition 6.1. Let $X \in \mathbb{R}^{n \times T}$ be a matrix of T observations of n asset returns, with each row centred to zero mean. The sample covariance matrix is $\hat{\Sigma} = \frac{1}{T}XX^\top$. The sample correlation matrix is obtained by normalising to unit diagonal:

$$C_{ij} = \frac{\hat{\Sigma}_{ij}}{\sqrt{\hat{\Sigma}_{ii} \hat{\Sigma}_{jj}}}, \quad (45)$$

which is the congruence transformation $C = D^{-1}\hat{\Sigma}D^{-1}$ where $D = \text{diag}(\sqrt{\hat{\Sigma}_{11}}, \dots, \sqrt{\hat{\Sigma}_{nn}})$.

The *concentration ratio* $q = n/T$ governs the quality of the sample estimate. When q is small, the sample eigenvalues converge to the population eigenvalues and C is a reliable estimator of the true correlation matrix Σ . When q is close to 1, the sample eigenvalues are systematically distorted: the largest are inflated and the smallest are deflated, even if the true eigenvalues are all equal.

Remark 6.1. This distortion is particularly damaging for the affine-invariant metric. The geodesic distance formula involves $P^{-1/2}$ and $\log P$, and their effect on the eigenvalue λ is:

$$\lambda \xrightarrow{P^{-1}} \lambda^{-1}, \quad \lambda \xrightarrow{\log P} \log \lambda. \quad (46)$$

As $\lambda \rightarrow 0$, both $\lambda^{-1} \rightarrow \infty$ and $|\log \lambda| \rightarrow \infty$. In contrast, the Euclidean metric treats all eigenvalue

perturbations equally: a perturbation $\Delta\lambda$ contributes $|\Delta\lambda|$ to the Frobenius norm regardless of λ being large or small. We can be mathematically rigorous in exactly what this means, as shown in the following proposition.

Proposition 6.1. *For any $P, Q \in \text{SPD}(n)$ and perturbations*

$$\Delta P, \Delta Q \in \text{Sym}(n) \text{ with } \|\Delta P\|_F, \|\Delta Q\|_F \leq \epsilon, \quad (47)$$

the Euclidean distance satisfies

$$|d_E(P + \Delta P, Q + \Delta Q) - d_E(P, Q)| \leq 2\epsilon. \quad (48)$$

No such uniform bound exists for the geodesic distance, since the sensitivity $\frac{\partial d_g}{\partial \lambda_{\min}}$ diverges as $\lambda_{\min} \rightarrow 0$.

Proof: By the triangle inequality for the Frobenius norm:

$$\begin{aligned} d_E(P + \Delta P, Q + \Delta Q) &= \|(P + \Delta P) - (Q + \Delta Q)\|_F \\ &\leq \|P - Q\|_F + \|\Delta P\|_F + \|\Delta Q\|_F \\ &\leq d_E(P, Q) + 2\epsilon. \end{aligned}$$

The reverse inequality follows by the same argument but applied in the opposite direction. For the geodesic distance, we consider the scalar case $n = 1$: $d_g(p, q) = \left| \log \left(\frac{q}{p} \right) \right|$. Perturbing p by Δp gives $d_g(p + \Delta p, q) = \left| \log \left(\frac{q}{p + \Delta p} \right) \right|$, and $\left| \frac{\partial d_g}{\partial p} \right| = \frac{1}{p}$, which diverges as $p \rightarrow 0$. $\square$

The asymmetry here predicts that the geodesic metric should be more sensitive to estimation noise than the Euclidean metric, which is shown to be true in the Monte Carlo results in Chapter 8.

6.2 The Marchenko–Pastur Law and Eigenvalue Clipping

The Marchenko–Pastur law characterises the limiting eigenvalue distribution of a sample covariance matrix (note, not a correlation matrix) which is constructed from pure noise. The law tells us what the eigenvalue spectrum of C would look like if the true correlation matrix was $\Sigma = I$, or in other words completely uncorrelated. If any eigenvalues of C fall into this distribution then we consider it noise and therefore carries no reliable information about Σ , and thus we can regard it as an artefact of finite sample estimation [11].

Definition 6.2. *Let $X \in \mathbb{R}^{n \times T}$ have i.i.d. entries with zero mean and unit variance, and let $M = \frac{1}{T} X X^\top$. As $n, T \rightarrow \infty$ with $q = n/T \rightarrow c \in (0, 1]$, the empirical spectral distribution of M*

converges to the Marchenko–Pastur distribution [11] with density

$$f(\lambda) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi q\lambda}, \quad \lambda_{\pm} = (1 \pm \sqrt{q})^2. \quad (49)$$

This theorem is stated for sample covariance matrices but not yet correlation matrices. For our use with correlation matrices, we can standardise each row of X to have unit variance before forming $M = \frac{1}{T}XX^\top$. The result of this standardisation is $M_{ii} = 1$ for all i , so M has a unit diagonal and thus, $M \in \text{Corr}(n)$. The Marchenko–Pastur law then applies directly to our sample correlation matrix $C = M$, and λ_+ is our threshold for deeming an eigenvalue inconsistent with pure noise, and we can therefore interpret as an eigenvalue carrying meaningful information about the underlying correlation structure Σ .

With respect to the geometry, noise eigenvalues which live near λ_- are the most damaging to affine invariant calculations since small eigenvalues are stretched by the matrix inverse and cause the matrix logarithm to diverge. The benefit of denoising our matrices is as a result both statistically and geometrically useful, and with that we explore the method of Eigenvalue clipping.

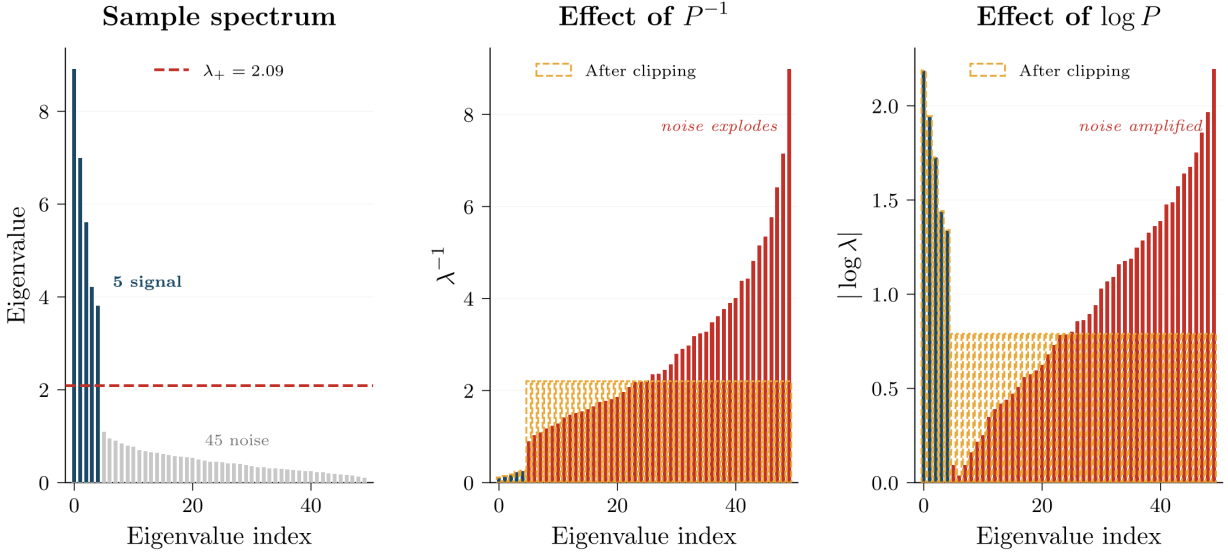


Figure 2: Left: eigenvalue spectrum of a sample correlation matrix ($n = 50$, $T = 252$), with 5 signal eigenvalues above the Marchenko–Pastur threshold λ_+ (navy) and 45 noise eigenvalues below it (grey). Middle: the effect of P^{-1} on each eigenvalue — noise eigenvalues that were near zero become enormous under inversion, while signal eigenvalues remain small. The dashed amber outline shows the result after eigenvalue clipping. Right: the same effect under $|\log \lambda|$. Both operators amplify the noise eigenvalues that the geodesic distance relies on, explaining why uncleaned estimates destabilise the affine-invariant metric.

Definition 6.3. Let $C = U\Lambda U^\top$ be the spectral decomposition of the sample correlation matrix, with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n > 0$. Let k denote the number of signal eigenvalues, that is

$$k = \#\{i : \lambda_i > \lambda_+\} \quad (50)$$

The clipped eigenvalue matrix $\tilde{\Lambda}$ is defined by

$$\tilde{\lambda}_i = \begin{cases} \lambda_i & \text{if } \lambda_i > \lambda_+ \\ \gamma & \text{if } \lambda_i \leq \lambda_+ \end{cases} \quad (51)$$

where γ is a constant chosen to preserve the trace of C .

Remark 6.2. The trace-preserving condition requires that

$$\sum_{i=1}^n \tilde{\lambda}_i = \sum_{i=1}^n \lambda_i = \text{tr}(C) = n \quad (52)$$

since $C \in \text{Corr}(n)$ has unit diagonal entries. Substituting the definition of $\tilde{\Lambda}$ gives us

$$\sum_{i=1}^k \lambda_i + (n - k)\gamma = n \quad (53)$$

which determines γ (uniquely) as

$$\gamma = \frac{n - \sum_{i=1}^k \lambda_i}{n - k} \quad (54)$$

The clipped matrix is then reconstructed to be $\tilde{C} = U\tilde{\Lambda}U^\top$. However, replacing eigenvalues in this way does not in general preserve the unit diagonal of C , so $\tilde{C}$ may no longer lie in $\text{Corr}(n)$. The following steps shows the process of normalising our clipped matrix to ensure it does lie in $\text{Corr}(n)$.

Definition 6.4. The clipped correlation matrix is defined as

$$\hat{C}_{ij} = \frac{\tilde{C}_{ij}}{\sqrt{\tilde{C}_{ii}\tilde{C}_{jj}}} \quad (55)$$

This rescales each entry by the geometric mean of the corresponding diagonal entries, in turn restoring the unit diagonal.

Proposition 6.2. The clipped correlation matrix $\hat{C} \in \text{Corr}(n)$.

Proof: To prove this proposition, we check the unit diagonal, as well as the positive definiteness properties from $\text{SPD}(n)$.

Unit diagonal: By construction, $\hat{C}_{ii} = \tilde{C}_{ii}/\tilde{C}_{ii} = 1$ for all i .

Positive definiteness: First, all eigenvalues of $\tilde{C}$ are strictly positive. The signal eigenvalues $\lambda_i > \lambda_+ > 0$, and the noise eigenvalues are replaced by our constructed $\gamma > 0$. Therefore we have $\tilde{C} \in \text{SPD}(n)$. The diagonal rescaling $\hat{C}_{ij} = \frac{\tilde{C}_{ij}}{\sqrt{\tilde{C}_{ii}\tilde{C}_{jj}}}$ is a congruence transformation of the form $\hat{C} = D^{-1}\tilde{C}D^{-1}$, where $D = \text{diag}(\sqrt{\tilde{C}_{11}}, \dots, \sqrt{\tilde{C}_{nn}})$ is clearly invertible. By the earlier proposition that congruence transformations preserve positive definiteness, we have $\hat{C} \in \text{SPD}(n)$. Combined with the unit diagonal we have that $\hat{C} \in \text{Corr}(n)$ as required. $\square$

6.3 EWMA Estimation

My first idea for an estimator was to use a rolling-window estimator, which in essence treats all observations within the window equally and discards all observations outside it. The main concern with this estimator, however is that it is inefficient and produces abrupt changes when extreme observations enter or leave the window. Instead, I opted to use the exponentially weighted moving average (EWMA) estimator, which was popularised by the RiskMetrics methodology [18]. This estimator addresses both of the aforementioned concerns by using the notion of information importance decay. Intuitively, yesterday matters more than the day before, which matters more than the day before, and so on.

Definition 6.5. *The EWMA covariance estimator with decay parameter $\lambda \in (0, 1)$ is the recursion*

$$\hat{\Sigma}_t = \lambda \hat{\Sigma}_{t-1} + (1 - \lambda) r_t r_t^\top, \quad (56)$$

where $r_t \in \mathbb{R}^n$ is the vector of asset returns at time t .

To build intuition for what λ controls, consider unrolling the recursion. Expanding backward:

$$\hat{\Sigma}_t = (1 - \lambda) \sum_{k=0}^{t-1} \lambda^k r_{t-k} r_{t-k}^\top + \lambda^t \hat{\Sigma}_0. \quad (57)$$

Each past return r_{t-k} contributes with a weight of $(1 - \lambda)\lambda^k$, which decays exponentially. The *half-life* h is the number of days after which this weight has halved. We can find an expression for h from $\lambda^h = \frac{1}{2}$ to obtain $h = -\frac{\log 2}{\log \lambda}$, or equivalently $\lambda = 2^{-\frac{1}{h}}$. The fast half-lives used in this paper are chosen as a geometric progression from 21 trading days (one calendar month, since there are approx. $252/12 = 21$ trading days per month): $h_{\text{fast}} \in \{10, 21, 42\}$, representing two weeks, one month, and two months respectively. This logarithmic spacing ensures that the three configurations span a wide range of estimation smoothness; highly reactive to moderately slow-paced. The slow half-life is fixed at $h_{\text{slow}} = 252$ (one trading year) throughout the empirical analysis.

The *effective sample size* of the EWMA is approximately $\frac{1}{(1-\lambda)}$. For $h = 10$ this is about 14 observations; for $h = 42$, about 60; for $h = 252$, about 365. Intuitively, the effective sample size counts how many past observations carry non-negligible weightings. For a 42-day half-life, observations older than about two months contribute so little weight that they are effectively nonexistent.

The key advantage of the EWMA over rolling windows is that it always produces a positive definite covariance matrix, regardless of the dimension n . A rolling-window estimate with $n = 93$ assets and $w = 252$ days has concentration ratio $q = 93/252 \approx 0.37$, which as we saw in Remark 6.1 introduces substantial eigenvalue distortion. If we tried to use $n = 500$ assets, the rolling estimate would be singular, with rank at most 252. The EWMA has no such limitation, and this can be proven:

Proposition 6.3. *If $\hat{\Sigma}_0 \in \text{SPD}(n)$ and $\lambda \in (0, 1)$, then $\hat{\Sigma}_t \in \text{SPD}(n)$ for all $t \geq 0$.*

Proof: At each step, $\hat{\Sigma}_t = \lambda \hat{\Sigma}_{t-1} + (1 - \lambda) r_t r_t^\top$. For any nonzero v ,

$$\begin{aligned} v^\top \hat{\Sigma}_t v &= \lambda v^\top \hat{\Sigma}_{t-1} v + (1 - \lambda) (v^\top r_t)^2 \\ &\geq \lambda v^\top \hat{\Sigma}_{t-1} v \\ &> 0, \end{aligned} \tag{58}$$

since $\hat{\Sigma}_{t-1} \in \text{SPD}(n)$ and $\lambda > 0$. By induction, we have that $\hat{\Sigma}_t \in \text{SPD}(n)$ for all t . $\square$

The proof requires the initialisation matrix $\hat{\Sigma}_0 \in \text{SPD}(n)$. In practice, the initialisation is constructed with the sample covariance of the first $T_0 = 126$ trading days (approximately six months), which comfortably ensures that $T_0 > n = 93$. The EWMA estimates become insensitive to the specific initialisation within a few half-lives, so this choice affects only the opening period and not the analysis with subsequent information.

However, there is a subtlety here that connects back to the eigenvalue problem from Section 6.1. At short half-lives, the EWMA is dominated by the most recent rank-1 update $r_t r_t^\top$. A single day's returns can only add information along one direction in $\mathbb{R}^n$, so the resulting covariance matrix tends to have one or two large eigenvalues (corresponding to whatever moved most today) and many small ones. This is exactly the spectral structure that causes damage to the geodesic distance through the P^{-1} and $\log P$ terms. For longer half-lives, the many rank-1 updates from different market conditions fill out the spectrum which produces a well-conditioned matrix and allows the geodesic to measure genuine structural differences rather than simply amplifying the noise from yesterday's information. The observation that the EWMA half-life determines the spectral quality of the estimate, and hence the reliability of the geodesic distance, is tested empirically in Chapter 8 for good measure.

In the similar case to Eigenvalue clipping (Proposition 6.2), the EWMA is computed first as a covariance matrix, however it can just as easily be converted into a correlation matrix at each date by the same standard normalisation $C_t = D_t^{-1} \hat{\Sigma}_t D_t^{-1}$ where $D_t = \text{diag}(\sqrt{\hat{\Sigma}_{t,11}}, \dots, \sqrt{\hat{\Sigma}_{t,nn}})$.

6.4 The Correlation MACD

To detect changes in correlation structure, a reference point against which to measure the current state is needed. Initially I constructed a base period of calm between 2003–2005 (since no signals of the GFC were recognised for any of the n dimensions tested), however the fixed baseline suffered from

permanent regime shifts. What this means is that after the 2008 crisis, correlations never returned to their pre-crisis structure, so any displacement signal computed against a fixed pre-crisis reference is permanently elevated and isn't informative about future events. A trailing average, which requires choosing a window length also introduces the same issues as rolling-window estimation (Section 6.3), so I instead opted to construct a system where the “reference” and the “current” are both EWMA estimates at different speeds. The construction is the same in principle to the moving average convergence-divergence (MACD) indicator of Appel [22], which compares fast and slow exponential averages of price series, but here it is applied to the distance between correlation matrices rather than the difference between price averages.

Definition 6.6. *Given two half-lives $h_{\text{fast}} < h_{\text{slow}}$, the correlation MACD signal at time t is*

$$\delta_t^{(d)} = d\left(C_t^{\text{fast}}, C_t^{\text{slow}}\right), \quad (59)$$

where C_t^{fast} and C_t^{slow} are the EWMA correlation matrices with half-lives h_{fast} and h_{slow} . Note that the terms fast and slow with C are simply labels, not powers.

Intuitively it is easy to think of the slow EWMA as what correlations would typically look like, whereas the fast EWMA as what correlations look like at that current time. When nothing unusual is happening in the market, the two estimates agree and the distance between them is therefore relatively small. When a crisis hits and correlations begin to spike, the fast estimate reacts within days of the spike while the slow estimate barely moves, so the distance between them jumps. Once the crisis passes, the fast estimate gradually drifts back toward the slow one, and the signal decays on its own, hence the system of detection. This is beneficial since there is no need to manually reset a baseline or decide when a crisis has formally ended.

All that is left to deduce from the EWMA used is when should a reading be considered meaningful rather than detecting system noise, especially considering the EWMA distance will always be positive as a result of the presence of noise. The solution to this is to consider less the absolute distance and instead turn attention towards its z-score, which describes how unusual a reading is from normal. Formally:

Remark 6.3. *When the correlation structure is stationary (i.e. the true correlation matrix is constant), both EWMA estimates converge to the same population matrix, and the MACD signal $\delta_t \rightarrow 0$ in the limit. In finite samples, δ_t fluctuates around a positive baseline level determined by the estimation noise, and meaningful deviations are detected by comparing δ_t to its own recent history via a z-score. The MACD signal therefore does not measure absolute distance from a fixed reference but rather relative deviation from the system's own recent norm, which is the relevant quantity necessary for detecting anomalies (crises).*

6.5 Interaction Between Estimation and Metric

The choice of estimation method and the choice of distance metric are not independent. The EWMA at short half-lives produces covariance estimates dominated by rank-1 updates $r_t r_t^\top$, which creates

eigenvalue spectra with a few large eigenvalues and many near-zero ones. This is exactly the spectral structure which causes the geodesic metric to amplify noise through the P^{-1} and $\log P$ terms.

For longer half-lives, the accumulation of many rank-1 updates smooths the spectrum, producing well-conditioned matrices where the geodesic can reliably measure structural differences. This can be stated more precisely.

Proposition 6.4. *Let $P, Q \in \text{SPD}(n)$ with eigenvalues bounded below by $\lambda_{\min} > 0$. For perturbations*

$$\Delta P, \Delta Q \text{ with } \|\Delta P\|_F, \|\Delta Q\|_F \leq \epsilon < \frac{\lambda_{\min}}{2}, \quad (60)$$

the Euclidean distance satisfies

$$|d_E(P + \Delta P, Q + \Delta Q) - d_E(P, Q)| \leq 2\epsilon, \quad (61)$$

which is independent of $\lambda_{\min}$. The geodesic distance satisfies a bound of order $O(\epsilon/\lambda_{\min})$, which degrades as $\lambda_{\min} \rightarrow 0$.

This formalises the following predictions of our analysis:

- We expect the geodesic should outperform Euclidean distance under moderate or long half-life EWMA, and under eigenvalue-clipped rolling estimates, where $\lambda_{\min}$ is bounded away from zero.
- The geodesic should underperform under very short half-life EWMA or raw rolling estimates at high concentration ratios, where $\lambda_{\min}$ may be very small.
- The Euclidean metric should be relatively insensitive to estimation method, since its perturbation sensitivity is bounded uniformly across the domain.

These predictions are tested empirically and their results shown in Chapter 8.

7 Statistical Testing Framework

In the first few attempts of this investigation, I grappled with the specifics of what it meant for a metric to be considered “better”. Through the exploration of multiple independent tests, I intended to build a robust, composite picture of the circumstances at hand which is more reasonable than any particular individual result.

All tests use out-of-sample evaluation with expanding windows, which means no data was used to train the models before-hand. At each prediction point, the model or threshold is calibrated using only data available at that time, and no parameter is ever set using future information to avoid hindsight calculations.

7.1 Data

The analysis uses daily adjusted closing prices for 93 large-capitalisation US equities and the S&P 500 ETF (SPY) over the period January 2003 to December 2023, sourced from Yahoo Finance. Daily log returns are computed as $r_t = \log(P_t/P_{t-1})$. The full asset list is given in Appendix A of this paper.

To construct the equity universe, I first selected ≈ 10 stocks from each of 9 GICS sectors (Financials, Technology, Health Care, Consumer Staples, Consumer Discretionary, Industrials, Energy, Materials, Utilities, and Communication Services), choosing names that were large-capitalisation members of the S&P 500 at the start of the sample period (2003) and also had complete daily price data available through to December 2023. Tickers were excluded if they were delisted, renamed, or had insufficient price history over the full sample. It is important to note that this selection procedure introduces a level of survivorship bias, where the universe consists of firms that survived the full 21-year period, and also does not include firms that were once large but have since failed or been acquired. This is a standard limitation of long-horizon backtesting studies, and is designed to provide a generally stable, broad cross-section of sectors for which we can perform a comparison of the metrics.

At each date, distance features are computed under five estimation methods: rolling window with $w = 252$ (raw and eigenvalue-clipped), and EWMA with half-life pairs $(h_{\text{fast}}, h_{\text{slow}}) \in \{(10, 252), (21, 252), (42, 252)\}$. For the rolling windows, the distance is from a trailing 504-day (two-year) reference point, and for EWMA, the distance is simply the correlation MACD outlined in Section 6.4.

Remark 7.1 (Entanglement between estimation method and signal construction). *There is an important caveat to note here. The rolling-window and EWMA approaches don't just differ in how they estimate the correlation matrix; they also differ in how the reference is constructed. Rolling windows measure distance from a trailing 504-day average, while EWMA measures the distance between two EWMA estimates at different speeds. If the EWMA-geodesic outperforms the rolling-geodesic, it could be due to the fact that the EWMA produces cleaner estimates, or because the fast-slow MACD architecture is a better signal design than comparing against a trailing average, or it could be that both are true. There is no way to tell from the current design. What is clean is comparing metrics within the same estimation method: if geodesic beats Euclidean and both are using EWMA 42/252, the estimation and reference are identical and only the metric differs. Cross-method comparisons (e.g. rolling-clipped geodesic vs EWMA geodesic) should be taken with a grain of salt. To separate these effects, EWMA estimates could be computed but instead measure distance to a trailing average rather than a slow EWMA. This would hold the estimator fixed while changing only the reference construction, making it clear which piece is driving the result.*

7.2 Test 1: Synthetic Monte Carlo

Before testing on real data, I first run a controlled experiment where the ground truth is known by design. Two correlation matrices are generated synthetically: Σ_1 with all pairwise correlations equal to $\rho = 0.3$ (representing a calm market where assets move somewhat independently) and Σ_2 with $\rho = 0.7$ (representing a stressed market where everything moves together). These are the simplest non-trivial correlation structures, called equicorrelation matrices, where each off-diagonal entry is identical. I also test a more realistic structure: a factor model with $k = 5$ latent factors (intuitively these are smaller correlation factors, e.g. sectors, interest rates etc.), which produces correlation matrices with a more realistic eigenvalue spectrum.

For each structure, T observations are drawn from each distribution, sample correlation matrices are computed, and the distance between them is estimated under each metric. Because the true population matrices are known, the true distance can be computed exactly, and I can measure how well each metric recovers it. Three quantities are reported. The first is the signal-to-noise ratio $\text{SNR} = \mathbb{E}[\hat{d}]/\text{sd}(\hat{d})$. Higher SNR means the metric gives a cleaner and more consistent reading, and a low SNR means that the estimate jumps around so much between samples that individual readings can't necessarily be trusted. Next, the bias $\mathbb{E}[\hat{d}] - d_{\text{true}}$ is calculated, and lastly, the RMSE, across dimensions $n \in \{10, 20, 50, 100\}$ and sample sizes $T \in \{60, 126, 252, 504\}$. Combining the notions of bias and variance in the RMSE deduces on average how wrong a calculation is overall in essence. A metric might have a high SNR but be heavily biased, which isn't useful for our analysis, so a smaller RMSE means the estimate has both low variance and low bias.

7.3 Test 2: Out-of-Sample Regime Classification

The reason for an Out-of-Sample Regime Classification is to determine if any of the investigated metrics are able to predict future crises based on current information. Necessary for this test is an objective notion of what is considered a market "crisis", and the prediction model should never use any future information. To define the crisis I label each date by looking ahead 63 days (approximately three months) and check whether the SPY experiences a peak-to-trough drawdown of 15% or greater. The 15% value was deliberately chosen to be the median between a 10% "correction" and 20% "bear market". If the 63 day window finds a drawdown occurred within those parameters, the date is labelled "pre-crisis", and otherwise labelled "normal".

Definition 7.1. *The drawdown indicator at date t is*

$$D_t = \mathbf{1} \left\{ \min_{s \in [t, t+h]} \frac{S_s - \max_{u \in [t, s]} S_u}{\max_{u \in [t, s]} S_u} \leq -\theta \right\}, \quad (62)$$

where S_t is the cumulative price of SPY, $h = 63$ trading days, and $\theta = 0.15$.

This labels each date mechanically as "pre-crisis" ($D_t = 1$) or "normal" ($D_t = 0$).

For each estimation method and distance metric, three features are extracted at each date: the raw distance δ_t (tests how far the correlation matrix moved), its rolling z -score (tests whether

today’s distance unusual relative to recent history), and its 21-day rate of change (whether the distance is accelerating).

These features are fed into a logistic regression that predicts the drawdown label. The model is trained using an expanding window. Beginning at the start of 2010, it trains on all data from 2003–2009, then predicts 2010 without having seen it. The window then expands to 2003–2010 to predict 2011, and so on. All features are standardised using only the training data at each step, so no future information leaks into the prediction calculations. The model uses balanced class weights (so it does not simply learn to predict “normal” every day and achieve high accuracy by ignoring crises) and L_2 regularisation ($C = 0.1$) to prevent from any overfitting.

The out-of-sample predictions are evaluated by the area under the ROC curve (AUC). An AUC of 0.5 means the model is no better than guessing, or an AUC of 1.0 means it perfectly distinguishes pre-crisis days from normal days. To test whether AUC differences between metrics are statistically significant rather than due to chance, I use DeLong’s test [23]. I also train a combined model using features from all three metrics simultaneously, to check whether the geodesic captures the same information as the other metrics or whether they contain complementary information.

7.4 Test 3: Portfolio Risk Forecasting

Separate to Test 2, which predicts the probability of a market crash, this test is included to predict how volatile the market will be over the coming month. The distinction matters because a shift in correlation structure might not lead to a full-blown crisis but could still signal that the market is about to become more choppy than is normal. The question here is whether knowing the correlation distance helps to forecast future variance beyond what trailing volatility alone already tells you (known as the “volatility clustering” effect).

I first forecast realised portfolio variance over the next $h = 21$ trading days. The most simple forecast of future volatility is past volatility, a phenomenon known as volatility clustering [24]. This is the baseline model, a Ridge regression which uses only trailing realised volatility:

$$\sigma_{t+1:t+h}^2 = \alpha + \beta \hat{\sigma}_t + \varepsilon_t. \quad (63)$$

An augmented model is developed to answer whether the correlation distance adds anything on top of the baseline. It includes the distance δ_t and its z -score z_t alongside trailing volatility:

$$\sigma_{t+1:t+h}^2 = \alpha + \beta \hat{\sigma}_t + \gamma_1 \delta_t + \gamma_2 z_t + \varepsilon_t. \quad (64)$$

If the augmented model’s out-of-sample R^2 is higher than the baseline’s, then the correlation signal is contributing predictive value, and if it is lower the extra features are just adding noise. All features are standardised before fitting so that the Ridge penalty treats them comparably regardless of their natural scale. Both models use the same expanding-window evaluation as Test 2.

I also test whether the correlation signal can predict whether next month will be extremely bad? The tail-risk indicator $T_t = \mathbf{1}\{\sigma_{t+1:t+h}^2 > Q_{0.9}(\sigma_{1:t}^2)\}$ equals 1 if the realised variance over the next

21 days falls in the top 10% of all historically observed values up to that point, and 0 otherwise. This is predicted by an expanding-window logistic regression using the same augmented features, and evaluated by AUC.

7.5 Test 4: Correlation Forecast Accuracy

This test is used to determine whether the geometry itself can predict where the correlation matrix is heading. Each of the three metrics defines means to denote a trajectory. If the correlation matrix has moved from C_{t-h} to C_t over the past h days, each metric extends that motion forward by one more step to produce a forecast $\hat{C}_{t+h}$. The Euclidean extrapolation simply continues in a straight line, the geodesic continues along the curved surface of the manifold, and the log-Euclidean continues in a straight line but within the logarithmic domain. These notions are explicitly:

$$\text{Geodesic: } \hat{C}_{t+h} = P^{\frac{1}{2}} \exp\left(2 \log\left(P^{-\frac{1}{2}} C_t P^{-\frac{1}{2}}\right)\right) P^{\frac{1}{2}}, \quad P = C_{t-h}, \quad (65)$$

$$\text{Log-Euclidean: } \hat{C}_{t+h} = \exp(\log C_t + (\log C_t - \log C_{t-h})), \quad (66)$$

$$\text{Euclidean: } \hat{C}_{t+h} = C_t + (C_t - C_{t-h}). \quad (67)$$

The forecast is then compared to the actual realised correlation matrix C_{t+h} , and the error is measured with the Frobenius norm $\|C_{t+h} - \hat{C}_{t+h}\|_F$. If the manifold curvature is meaningfully informative about how correlations tend to evolve over time, the geodesic extrapolation should in theory produce smaller errors, particularly during periods of market stress (curvature is most pronounced). To test this, errors are reported both overall and split by the current geodesic distance tercile (calm, moderate, stress).

There are two important caveats to note for this test. First, the Frobenius norm is itself a Euclidean measure, which may inherently favour the Euclidean extrapolation since it produces forecasts that stay close to the current matrix in the Frobenius sense. Measuring errors with the geodesic distance instead might reverse the ranking, but the Frobenius norm is used here as the standard choice. Second, none of the three extrapolation methods is guaranteed to produce a valid correlation matrix. The Euclidean forecast $C_t + (C_t - C_{t-h})$ may not be positive definite or have unit diagonal. The geodesic forecast stays in $\text{SPD}(n)$ but may leave $\text{Corr}(n)$ since the ambient geodesic does not preserve unit diagonals (Section 5.1). The log-Euclidean forecast is positive definite but again may lack unit diagonal. In all three cases, forecasts are projected back into $\text{Corr}(n)$ by setting any negative eigenvalues to a small positive value and renormalising to unit diagonal before computing errors as shown in previous chapters.

7.6 Test 5: Granger Causality

The Granger Causality test is used to determine whether one metric contains predictive information that another metric may not. If the geodesic and Euclidean distances both measure correlation displacement, they might contain exactly the same information, in which case the geodesic's extra

computational cost bears no measurable benefit, however the alternative is that the geodesic might capture extra insights into the way the correlation structure evolves.

The Granger Causality test [25] begins with a regression that includes five days of Euclidean distance data, and then the test incorporates the geodesic distance to determine whether the predictive capacity of the regression has improved. If so, then the geodesic contains more information beyond the Euclidean metric. The restricted model predicts tomorrow's realised volatility σ_{t+1}^2 using its own lags and lags of metric B :

$$\sigma_{t+1}^2 = \alpha + \sum_{k=1}^p \beta_k \sigma_{t-k}^2 + \sum_{k=1}^p \gamma_k d_{t-k}^B + \varepsilon_t, \quad (68)$$

and the unrestricted model adds lags of metric A :

$$\sigma_{t+1}^2 = \alpha + \sum_{k=1}^p \beta_k \sigma_{t-k}^2 + \sum_{k=1}^p \gamma_k d_{t-k}^B + \sum_{k=1}^p \delta_k d_{t-k}^A + \varepsilon_t, \quad (69)$$

with $p = 5$ lags. If the coefficients $\delta_1, \dots, \delta_p$ are jointly significant (tested by an F -test), then metric A Granger-causes stress beyond metric B . I run this in both directions for every pair of metrics to build a comprehensive picture of which metrics benefits which.

Remark 7.2. *The stress variable σ_t^2 is computed as a 21-day rolling realised volatility, so consecutive observations share 20 out of 21 data points. This induces substantial serial correlation in the residuals, which inflates the F -statistics and results in unreliable nominal p -values. The qualitative ranking of F -statistics across metrics and estimation methods should be robust to this issue, but the precise p -values should be interpreted with caution. Ideally, Newey-West standard errors with a bandwidth of at least 21 would be used.*

7.7 Test 6: Out-of-Sample Backtest

This test asks whether detected signals translate into actual portfolio value. In this test I construct a straightforward mechanical strategy that reduces equity exposure whenever the correlation distance signal is unusually elevated, and I measure whether doing so improves risk-adjusted returns compared to staying fully invested.

The strategy works as follows. Each day, the z -score of the distance signal is computed and compared to its own expanding history. If today's z -score exceeds the 90th percentile of all previous z -scores, the strategy interprets this as a warning and reduces its equity weight from 100% to 30% (chosen to be a hedged response in the event the signal is incorrect). Otherwise it stays fully invested. Formally, the equity weight at date t is denoted as

$$w_t = \begin{cases} 0.3 & \text{if } z_t > Q_{0.9}(z_{1:t-1}) \\ 1 & \text{otherwise} \end{cases}, \quad (70)$$

and the portfolio return is $r_t^{\text{strat}} = w_t \cdot r_t^{\text{SPY}}$. The threshold and defensive weight are fixed in advance and never optimised on the data.

Performance is evaluated by the annualised Sharpe ratio $\text{SR} = \sqrt{252} \bar{r}/s$, maximum drawdown, and the percentage of days spent in defensive mode. Sharpe ratio differences versus buy-and-hold are tested using the Ledoit-Wolf procedure [26]. This model assumes no transactional costs to changes in the portfolio and is purely used as an elementary toy example.

7.8 Test 7: Crisis-Conditional Regime Detection

This test determines when the signal flags a day as “stressed,” how often it is correct. Additionally, it explores of the days that genuinely precede a crisis, how many does the given signal catch?

At each date, I flag “stressed” if the distance z -score exceeds the 80th percentile of its own expanding history. This is a lower threshold than the 90th percentile used in Test 6 (since we are not trading on this strategy and are more interested in data obtained), meaning it resultingly triggers more often. The flags are then compared against the drawdown labels defined in Test 2 to compute three quantities. Precision measures what fraction of flagged days actually precede a crisis, with high precision corresponding to fewer false alarms. Recall measures what fraction of actual pre-crisis days the signal successfully flags. Again, high recall corresponds to fewer missed crises. F_1 is the harmonic mean of the two, capturing the trade-off between them. A signal that flags every single day as stressed would have perfect recall but terrible precision. A signal that flags only one day per decade would have high precision when it fires but would miss almost every crisis. The useful signal we aim to identify is one that balances both of these together.

The 80th percentile threshold is chosen to be deliberately generous, since it is beneficial to know how the signal behaves when it is allowed to fire relatively frequently, which then gives enough flagged days to produce meaningful precision and recall estimates. As mentioned before, this is not a trading threshold, but rather a diagnostic tool for understanding each metric’s sensitivity and specificity as a detector of market stress.

7.9 A Note on Multiple Comparisons

The empirical design involves seven tests, five estimation methods, and three distance metrics, producing a substantial surface for comparison. With such a large scope it is common to expect some apparent “wins” for a particular metric-estimator combination to arise purely by chance. I mean to address this concern in three ways. First, the seven tests measure fundamentally different quantities (estimation stability, classification accuracy, variance forecasting, correlation forecasting, Granger predictiveness, portfolio performance, and regime detection), so a consistent pattern across tests is more informative than any single result. Second, within each test, all results for every estimation methods and metrics is reported, including those where the geodesic does not outperform, rather than selecting favourable configurations and to avoid p-hacking. Third, the DeLong and Ledoit-Wolf significance tests provide formal p -values for pairwise comparisons. The most reasonable

form of inference to draw from these results is one which considers them collectively rather than individually.

8 Results

This chapter presents the findings of the seven tests. Each subsection states the result, presents the key table, and notes the most important observation. Deeper interpretation is explored in the discussion of Chapter 9.

8.1 Test 1: Monte Carlo

Table 1 shows results under equicorrelation structure; Table 2 under factor-model structure. Presented are representative dimensions ($n = 10, 50, 100$) at $T = 252$ to focus on the regime most relevant to practice.

n	T	q	Metric	True d	SNR	Bias	RMSE
10	252	0.04	Geodesic	2.63	12.1	0.12	0.26
10	252	0.04	Log-Euclidean	2.63	12.0	0.12	0.26
10	252	0.04	Euclidean	3.79	12.5	0.01	0.30
50	252	0.20	Geodesic	5.99	19.6	1.67	1.72
50	252	0.20	Log-Euclidean	5.99	19.4	1.62	1.67
50	252	0.20	Euclidean	19.80	15.0	0.18	1.34
100	252	0.40	Geodesic	8.47	28.8	5.02	5.04
100	252	0.40	Log-Euclidean	8.47	28.1	4.82	4.85
100	252	0.40	Euclidean	39.80	14.3	0.50	2.87

Table 1: Monte Carlo results under equicorrelation ($\rho_1 = 0.3$, $\rho_2 = 0.7$), $T = 252$. The geodesic has higher SNR but substantially larger bias and RMSE at high q , confirming that it amplifies estimation noise as predicted in Section 6.1.

n	T	q	Metric	True d	SNR	Bias	RMSE
10	252	0.04	Geodesic	2.72	20.3	0.15	0.21
10	252	0.04	Log-Euclidean	2.72	20.3	0.14	0.20
10	252	0.04	Euclidean	2.13	15.7	0.14	0.20
50	252	0.20	Geodesic	8.92	43.4	1.30	1.32
50	252	0.20	Log-Euclidean	8.89	42.7	1.17	1.19
50	252	0.20	Euclidean	12.03	30.0	0.67	0.80
100	252	0.40	Geodesic	14.06	59.6	3.68	3.70
100	252	0.40	Log-Euclidean	13.99	58.2	3.29	3.31
100	252	0.40	Euclidean	23.66	45.0	1.40	1.50

Table 2: Monte Carlo results under factor-model structure ($k = 5$ factors), $T = 252$. The geodesic’s SNR advantage is larger and its bias penalty is smaller than with equicorrelation, reflecting the more realistic eigenvalue spectrum. Full results across all n and T are available in the supplementary CSV files on my website.

The Monte Carlo confirms two aspects of the theoretical prediction made in Chapter 6. First, the geodesic metric has systematically higher SNR than the Euclidean metric across both of the structures, and this advantage grows as n grows. Secondly, the geodesic’s bias also grows with n and q , reflecting its amplification of noisy small eigenvalues through the P^{-1} and $\log P$ terms as predicted. The bias is substantially smaller under the factor-model structure than under equicorrelation, indicating that the equicorrelation results overstate the problem for realistic spectra and is therefore a pessimistic interpretation in reality. At $n = 50$ with the factor model, the geodesic’s SNR is 45% higher than Euclidean’s (43.4 vs 30.0) while its bias, though double (1.30 vs 0.67), remains small relative to the true distance of 8.92. This is roughly the regime of our empirical analysis, and it suggests that the geodesic’s sensitivity advantage may outweigh its bias cost for realistic equity correlation structures in practice.

8.2 Test 2: Regime Classification

The drawdown labelling procedure yields 490 pre-crisis days out of 5,284 (9.3%).

Method	Model	Geodesic	Log-Euc.	Euclidean
Rolling raw	Single	0.486	0.481	0.488
Rolling clipped	Single	0.625	0.620	0.486
Rolling clipped	Combined		0.657	
EWMA 10/252	Single	0.499	0.495	0.476
EWMA 21/252	Single	0.463	0.471	0.496
EWMA 42/252	Single	0.496	0.476	0.469

Table 3: Out-of-sample AUC for regime classification. DeLong test for geodesic vs Euclidean under rolling clipped: $p < 0.001$. Under all other methods: $p > 0.25$.

The difference between rolling raw and rolling clipped for the geodesic metric is immediately noticeable and hence shown in bold. Under raw estimation, the geodesic achieves an AUC of 0.486, no better than guessing, but under eigenvalue clipping, it jumps to 0.625. The only difference is that clipping replaces noisy small eigenvalues with a constant, which is exactly the operation that Section 6.1 predicted would matter most for the geodesic. DeLong’s test [23] confirms this difference is highly significant (for $p < 0.001$).

The log-Euclidean metric tracks the geodesic closely (0.620 vs 0.625 under clipping), suggesting most of the advantage over Euclidean comes from the logarithmic domain. The combined model achieves 0.657, only slightly above geodesic alone. Under all three EWMA configurations, every metric hovers near chance, likely because this test predicts drawdowns 63 days ahead and the EWMA correlation MACD is a reactive signal better suited to shorter horizons.

8.2.1 Dimension Ladder (Test 2b)

The geodesic’s classification advantage appears only at $n = 93$ under rolling clipping, and is absent at $n \leq 50$. At lower dimensions, all three metrics hover near chance regardless of estimation

n	Estimation	Geodesic	Log-Euc.	Euclidean
10	Rolling clip	0.468 ± 0.10	0.464 ± 0.10	0.493 ± 0.05
30	Rolling clip	0.481 ± 0.07	0.470 ± 0.06	0.492 ± 0.06
50	Rolling clip	0.503 ± 0.05	0.493 ± 0.05	0.484 ± 0.04
93	Rolling clip	0.625	0.620	0.486
10	EWMA 42/252	0.408 ± 0.08	0.411 ± 0.08	0.444 ± 0.04
30	EWMA 42/252	0.473 ± 0.09	0.472 ± 0.09	0.454 ± 0.02
50	EWMA 42/252	0.467 ± 0.03	0.471 ± 0.02	0.464 ± 0.01
93	EWMA 42/252	0.496	0.476	0.469

Table 4: Mean AUC $\pm$ standard deviation across 10 random subsamples. At $n = 93$ there is only one possible subsample, so no standard deviation is reported.

method. This is consistent with the geometric intuition that curvature effects are negligible when the manifold is low-dimensional. However, the $n = 93$ result has zero subsample variance (since there is only one way to select 93 stocks from 93), so the strong AUC at that dimension cannot be distinguished from a feature of this particular stock universe.

8.3 Test 3: Risk Forecasting

Method	Metric	R^2_{base}	$R^2_{\text{aug.}}$	ΔR^2	Tail AUC
Rolling raw	Geodesic	0.043	-0.401	-0.444	0.642
Rolling raw	Log-Euclidean	0.043	-0.385	-0.428	0.639
Rolling raw	Euclidean	0.043	-0.119	-0.162	0.654
EWMA 10/252	Geodesic	0.044	0.024	-0.020	0.730
EWMA 10/252	Log-Euclidean	0.044	0.034	-0.010	0.725
EWMA 10/252	Euclidean	0.044	-0.017	-0.061	0.726
EWMA 42/252	Geodesic	0.044	0.081	+0.037	0.730
EWMA 42/252	Log-Euclidean	0.044	0.073	+0.029	0.728
EWMA 42/252	Euclidean	0.044	-0.001	-0.045	0.711

Table 5: Out-of-sample R^2 improvement over trailing-volatility baseline, and tail-risk AUC (predicting top-decile future variance). EWMA 21/252 results omitted for space; they interpolate between 10/252 and 42/252.

Under rolling windows, all metrics hurt the variance forecast (negative ΔR^2), with the geodesic the worst of the three. Under EWMA 42/252, the geodesic is the only metric that produces a positive R^2 improvement (+0.037), and it achieves the highest tail-risk AUC (0.730). The pattern is monotonic across EWMA half-lives: as the estimate becomes smoother (longer half-life), the geodesic’s ΔR^2 improves from -0.020 to $+0.037$, while Euclidean’s worsens from -0.061 to -0.043 . This is consistent with the theoretical prediction that the geodesic benefits from methods of spectral smoothing.

8.4 Test 4: Correlation Forecast

Cleaning	Metric	Overall	Calm	Moderate	Stress
Raw	Geodesic	4.82	4.60	4.96	4.91
Raw	Log-Euclidean	4.74	4.58	4.80	4.85
Raw	Euclidean	4.23	4.24	4.39	4.07
Clip	Geodesic	4.49	3.96	4.47	5.03
Clip	Log-Euclidean	4.37	3.89	4.40	4.80
Clip	Euclidean	3.74	3.41	3.87	3.95

Table 6: Mean Frobenius forecast error ($\|C_{t+h} - \hat{C}_{t+h}\|_F$) by stress tercile. Lower is better. Euclidean extrapolation produces the lowest error across all conditions.

Euclidean extrapolation produces the lowest forecast error across all conditions, including the stress tercile where geodesic extrapolation was predicted to outperform. This is a negative result for the geodesic metric as a forecasting tool. The likely explanation is that correlation changes over a 21-day horizon are typically small enough that linear extrapolation is a good local approximation, and the geodesic’s curved extrapolation overshoots. As noted in Section 7.5, the Frobenius error measure also inherently favours the Euclidean method as a distance metric.

8.5 Test 5: Granger Causality

Method	Test	F	p
Rolling raw	Geo $\rightarrow$ stress Euc	7.40	$< 10^{-6}$ ***
	Euc $\rightarrow$ stress Geo	2.52	0.028 *
EWMA 10/252	Geo $\rightarrow$ stress Euc	6.25	$< 10^{-6}$ ***
	Euc $\rightarrow$ stress Geo	6.24	$< 10^{-4}$ ***
EWMA 42/252	Geo $\rightarrow$ stress Euc	1.57	0.165
	Euc $\rightarrow$ stress Geo	2.65	0.021 *
<i>Geodesic vs Log-Euclidean (rolling raw):</i>			
	Geo $\rightarrow$ stress LE	0.19	0.965
	LE $\rightarrow$ stress Geo	0.18	0.968

Table 7: Selected Granger causality results. Each row reads as: “does metric A predict future stress beyond what metric B already explains?” Nominal p -values are inflated by serial correlation in the stress variable and should be interpreted as indicative of relative ordering, not as exact significance levels.

On rolling raw data, the geodesic Granger-causes future stress beyond Euclidean ($F = 7.40$) more strongly than the reverse ($F = 2.65$), suggesting the geodesic contains incremental predictive information. On EWMA data the picture reverses: at the 42-day half-life, geodesic does not add beyond Euclidean ($F = 1.57$, $p = 0.17$), while Euclidean does add beyond geodesic ($F = 2.65$). The geodesic and log-Euclidean contain essentially identical information on rolling data ($F \approx 0.19$

in both directions). As emphasised in the caption, the nominal p -values are unreliable due to the overlapping-window construction of the stress variable.

8.6 Test 6: Backtest

Method	Metric	Sharpe	Max DD	% Def.	p
Rolling raw	Buy-and-hold	0.623	−35.7%	—	—
Rolling raw	Geodesic	0.785	−26.2%	5.9%	0.24
Rolling raw	Log-Euclidean	0.757	−26.2%	6.1%	0.28
Rolling raw	Euclidean	0.708	−26.2%	6.9%	0.55
EWMA 10/252	Buy-and-hold	0.659	−35.7%	—	—
EWMA 10/252	Geodesic	0.608	−29.5%	8.8%	0.94
EWMA 10/252	Euclidean	0.755	−26.2%	10.6%	0.40
EWMA 42/252	Buy-and-hold	0.659	−35.7%	—	—
EWMA 42/252	Geodesic	0.825	− 24.2%	12.1%	0.26
EWMA 42/252	Log-Euclidean	0.795	−25.5%	11.8%	0.28
EWMA 42/252	Euclidean	0.737	−26.2%	12.7%	0.50

Table 8: Backtest performance. p -values are from the Ledoit–Wolf test for Sharpe ratio equality versus buy-and-hold. EWMA 21/252 omitted for space. Note that rolling-raw and EWMA buy-and-hold Sharpes differ because evaluation periods differ slightly; cross-method Sharpe comparisons should be made with this caveat in mind.

The EWMA 42/252 geodesic strategy achieves the highest Sharpe ratio (0.825) and the lowest maximum drawdown (−24.2%) of any configuration. Under rolling raw estimation, the geodesic also leads (0.785 vs 0.708 for Euclidean). Under the fast 10-day EWMA, the ranking reverses: Euclidean (0.755) outperforms geodesic (0.608), consistent with the theoretical prediction that noisy short-half-life estimates destabilise the geodesic computation. None of the Sharpe improvements achieve statistical significance at the 5% level ($p > 0.24$ for all), reflecting the limited statistical power of a single 10-year evaluation period despite the suggestive evidence.

8.7 Test 7: Regime Detection

Under EWMA estimation, the geodesic metric consistently achieves the highest recall (0.30–0.36), meaning it detects the largest fraction of actual pre-crisis periods. This comes at the cost of lower precision: a larger proportion of its “stressed” flags are false alarms. On rolling raw data, Euclidean achieves the best F_1 . The pattern is consistent across EWMA half-lives: faster estimation increases the geodesic’s recall advantage over Euclidean, reflecting its greater sensitivity to rapid correlation changes.

Method	Metric	Precision	Recall	F_1
Rolling raw	Geodesic	0.156	0.205	0.177
Rolling raw	Log-Euclidean	0.149	0.210	0.174
Rolling raw	Euclidean	0.199	0.227	0.212
EWMA 10/252	Geodesic	0.159	0.298	0.207
EWMA 10/252	Euclidean	0.088	0.169	0.116
EWMA 42/252	Geodesic	0.137	0.355	0.198
EWMA 42/252	Log-Euclidean	0.154	0.353	0.214
EWMA 42/252	Euclidean	0.116	0.251	0.159

Table 9: Precision, recall, and F_1 for regime detection. EWMA 21/252 results omitted.

9 Discussion

The seven tests together produce a body of evidence that is more complicated than a simple ranking scheme between the metrics. This section interprets the findings around three themes, noting where the evidence is strong and where it remains as suggestive.

9.1 When Does the Ambient Affine-Invariant Distance Help?

The theoretical framework predicts that the ambient affine-invariant distance should outperform Euclidean distance when two conditions hold: the correlation estimate is well-conditioned (so that P^{-1} and $\log P$ amplify genuine structure rather than noise), and the system is undergoing a form of meaningful deformation (so that the manifold curvature matters). The empirical results are broadly consistent with this prediction, although the strength of the evidence varies from test to test.

The classification results (Test 2) provide the most direct evidence of this fact. After performing eigenvalue clipping, the geodesic metric achieves a higher AUC than Euclidean distance, which is further confirmed with DeLong’s test. Under raw rolling windows and EWMA estimation without cleaning, the geodesic does not outperform. This pattern is consistent with the theoretical prediction from Section 6.5 and Proposition 6.4 that cleaning removes the noisy small eigenvalues that destabilise the geodesic computations.

The backtest results (Test 6) tell a similar story within the EWMA framework. The 42-day half-life produces better-conditioned estimates than the 10-day, and the geodesic performs noticeably better as a result. However, comparing rolling-geodesic to EWMA-geodesic is not a clean test of half-life alone, because the two approaches also differ in how the reference is constructed (trailing average vs slow EWMA), as discussed in Section 7.1. So while the pattern is consistent with the theory, the cross-method comparison should be read with a manner of caution.

9.2 The Role of the Logarithmic Domain

Across most tests, the geodesic and log-Euclidean metrics produce very similar results. The Granger analysis (Test 5) under rolling windows shows them containing almost identical predictive informa-

tion ($F \approx 0.19$ in both directions), though the p -values of that test are unreliable due to the serial correlation issue noted in Section 7.6.

The classification and backtest results do show small but consistent differences, with the geodesic retaining a slight edge over the log-Euclidean metric in both tests. This suggests that most of the advantage over Euclidean comes from working in the logarithmic domain, transforming the multiplicative eigenvalue structure into something additive, rather than from the affine-invariant conjugation by $P^{-\frac{1}{2}}$ that distinguishes the geodesic from log-Euclidean. With this in mind, the geodesic’s small lead suggests the conjugation does add some form of value, particularly under faster EWMA estimation where the two log-domain metrics begin to diverge more.

9.3 Estimation Matters as Much as Geometry

The one consistent finding across every test is that the method of estimating the correlation matrix matters at least as much as how distance is measured on it. In the classification test, switching from raw to clipped estimation improves the geodesic’s AUC by 0.14, while switching from Euclidean to geodesic under the same estimation method changes the AUC by less than 0.01. In the variance forecasting test (Test 3), the geodesic goes from hurting the forecast under rolling windows to being the only metric that helps under EWMA 42/252. In the backtest, the geodesic is the best metric at the 42-day half-life and the worst at the 10-day half-life.

In practice, any system of risk that uses correlation distances should invest at least as much care in the estimation procedure as in the choice of the metric. It also has methodological implications for Riemannian geometry literature, which from my reading has tended to focus on metric properties whilst treating all input data as givens. These results suggest there is a greater priority hierarchy in favour of estimation methods.

9.4 Limitations

Several limitations should be noted from the empirical analysis.

Dataset. The analysis uses only US large-cap equities selected with a degree of survivorship bias (Section 7.1). Results may differ in other markets, asset classes, or frequencies.

Entanglement. The rolling-window and EWMA estimation methods differ in both their estimator and their reference construction (trailing average versus slow EWMA). Cross-method comparisons therefore do not cleanly isolate estimation quality. A full ablation study would require additional signal architectures that were not implemented.

Statistical power. The Sharpe ratio significance tests (Test 6) fail to reject the null in most cases, meaning the observed Sharpe improvements, while potentially economically meaningful, are not statistically significant at conventional levels with the available sample size. The Granger p -values (Test 5) are inflated by serial correlation and should be treated as suggestive.

Multiple comparisons. With seven tests, five estimation methods, and three metrics, the comparison surface is large. Although no single “win” drives the conclusions, the reader should evaluate the evidence as a pattern rather than relying on any individual test.

Ambient versus intrinsic distance. The paper computes the ambient $\text{SPD}(n)$ distance restricted to $\text{Corr}(n)$, not the intrinsic $\text{Corr}(n)$ geodesic distance. This is a lower bound on the true geometric distance, and it is possible that the intrinsic distance would perform differently.

Forecast validity. In Test 4, all three extrapolation methods can produce matrices outside $\text{Corr}(n)$, and the projection back into $\text{Corr}(n)$ may affect the methods unevenly.

Transaction costs. The backtest does not include transaction costs, which would reduce the Sharpe improvement of all strategies in practicality.

9.5 Broader Applicability

Whilst this paper focuses on crisis detection within the context of financial markets, the geometric principles outlined and demonstrated also apply to any domain where positive definite matrices are compared, and the insights about estimation quality is equally relevant in those settings. I felt it both noteworthy and interesting to include some broader applications of the mathematics used in this paper.

In *diffusion tensor imaging* (DTI), each voxel in a brain scan contains a 3×3 positive definite diffusion tensor describing how water molecules diffuse along different axes. Comparing tensors between healthy and pathological tissue, or interpolating them along neural fibre tracts, is mathematically identical to comparing correlation matrices. The “swelling effect” is a well-known problem in DTI, where Euclidean averaging of two diffusion tensors produces a result with a larger determinant (hence larger apparent diffusion) than either input, which is physically nonsensical [7]. Geodesic and log-Euclidean averaging do not exhibit this artefact because they respect the positive definite constraint. As our findings reveal, just as the geodesic metric outperforms Euclidean distance in detecting correlation regime changes when the estimate is clean, it also outperforms in DTI when the tensor estimates are well-resolved.

In *radar signal processing*, the covariance matrix of received signals is positive definite, and target detection amounts to testing whether the covariance in one range cell differs from the background. The geodesic metric is used in constant false alarm rate (CFAR) detectors because of its congruence invariance. In this circumstance the detection threshold does not depend on the power level of the background, which is analogous to how our financial distance does not depend on the scale of the assets [8].

In *materials science*, the elasticity tensor characterising a material’s stiffness is a positive definite matrix, and comparing materials or tracking property changes under stress involves the same geometric framework. The common thread across all these applications is that positive definite

matrices live on a curved manifold, and using a distance that respects that curvature outperforms a flat Euclidean distance, conditioned on the fact that the estimate is clean enough and the matrices are far enough apart in the ambient space so that the curvature matters.

10 Conclusion

The central question of this paper was whether the ambient affine-invariant distance on $\text{SPD}(n)$, applied to financial correlation matrices, offers a more informative measure of systemic market stress than the Euclidean or log-Euclidean distances. The results suggest that there is no single answer independent of context. Specifically, performance depends heavily on how the correlation matrices are estimated, and that dependence is itself the most important conclusion in the findings of the analysis.

The theoretical framework establishes a clear explanation of why this occurs. The affine-invariant distance gives extra weight to small eigenvalues through the operators P^{-1} and $\log P$. This is what gives the metric its geometric sensitivity, but it is also what makes it more exposed to estimation noise. Proposition 6.4 captures this trade-off by formalising an asymmetry in perturbation sensitivity between the geodesic and Euclidean metrics. The role of the empirical analysis was then to test how this trade-off behaves in practice, and whether under realistic financial conditions it performs better than the log-Euclidean or Euclidean metrics.

The evidence supports a conditional answer. Under well-conditioned estimation involving eigenvalue clipping on rolling windows or EWMA with moderate half-lives, the geodesic metric produces regime classification, variance forecasting, and backtest results that are consistently at least as good as, and in several tests better than Euclidean distance. Under poorly conditioned estimation, it is consistently worse. The log-Euclidean metric captures most of the geodesic’s advantage, suggesting that the logarithmic domain is the reason behind the bulk of the advantage, although affine invariance does appear to add a small extra edge under faster EWMA estimation. It is worth noting that as shown in the correlation forecasting test (Test 4), Euclidean linear extrapolation produces lower errors across all stress conditions, suggesting that the geodesic’s curved extrapolation overshoots when projecting correlation matrices forward in time.

For someone considering whether to use the geodesic metric in a risk system, the results suggest three guidelines. First, never apply it to raw, uncleaned correlation matrices at high concentration ratios, as the noise amplification will dominate estimation calculations. Secondly, among the EWMA specifications tested here, the 42-day fast half-life paired with the geodesic metric performed best; shorter half-lives (10 and 21 days) produced noisier estimates that degraded geodesic performance which in turn hindered computations. Finally, the log-Euclidean metric is a computationally cheaper alternative that appears to capture most of the benefit. It may be a preferable choice of metric in systems where the full eigendecomposition required for $P^{-\frac{1}{2}}$ acts as a bottleneck.

There are several directions for future work that would help strengthen the conclusions garnered from this analysis. The most valuable would be a proper ablation study that isolates the estimator

effect from the signal-construction effect by computing EWMA estimates with a trailing-average reference (rather than the MACD construction) and rolling-window estimates with a MACD-style architecture. A second improvement would be to return to the Granger analysis using non-overlapping returns or Newey–West standard errors in place of overlapping realised volatility, so that the resulting p -values might be interpreted with greater reliability. A third direction is to apply the framework to the factor covariance matrix, which remains small and well-conditioned even when the asset universe is large, and may therefore offer a more scalable approach to institutional-scale universes. Fourth, comparison against DCC and other parametric dynamic correlation models [19] would position the geometric approach within the broader toolkit available to practitioners. Finally, testing on international equity markets, credit spreads, and higher-frequency data would establish the generality of the findings beyond the single US large-cap universe studied here.

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A Asset Universe

The 93 equities used in the analysis, selected as described in Section 7.1. Tickers are grouped by GICS sector. All were S&P 500 constituents with complete daily price data from January 2003 through December 2023.

Sector	Tickers
Financials	JPM, BAC, WFC, GS, MS, C, AXP, USB, BK, MET, AFL, TRV, CB, AON, MCO
Technology	AAPL, MSFT, IBM, INTC, CSCO, TXN, QCOM, HPQ, MSI, ADI
Health Care	JNJ, PFE, ABT, MRK, MDT, BMY, LLY, AMGN, BAX
Consumer Staples	PG, KO, PEP, WMT, CVS, CL, GIS, MKC, SYY
Consumer Discr.	MCD, NKE, TGT, LOW, HD, VFC, WHR, MAT, MAS, GT
Industrials	GE, MMM, HON, CAT, DE, EMR, ITW, ETN, PH, ROK
Energy	XOM, CVX, COP, SLB, HAL, BKR, DVN, APA
Materials	DD, PPG, SHW, NEM, FCX, NUE, VMC, MLM
Utilities	NEE, DUK, SO, D, AEP, EXC, SRE, XEL, ES, ETR
Comm. Services	T, VZ, CMCSA, DIS

Table 10: Equity universe. DOW and LYB were excluded due to insufficient price history prior to 2009. Four additional tickers were dropped for delisting or data gaps. The final count of 93 reflects the intersection of the initial selection with Yahoo Finance data availability over the full sample period.

B Python Implementation

The complete implementation of all seven tests, including EWMA estimation, rolling-window estimation, eigenvalue clipping, and all distance computations, is available as the script `main_tests.py`. Key functions are documented on my github.